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基于物联网的停车场监控数据采集系统设计与软件实现

摘 要

随着城市交通压力的日益加剧，智能化停车场系统的开发成为缓解这一问题的重要途径。本文提出了一种基于物联网技术的停车场监控数据采集系统，旨在通过实时监控停车场的环境信息以及车位占用状况，提高停车效率与管理质量。该系统以高性能 STM32F103C8T6 单片机为核心控制单元，整合了空气温湿度传感器、液体传感器及红外传感器，用于精确监测停车场环境参数和车位使用情况。

综上所述，本研究开发的基于物联网的停车场监控数据采集系统不仅提高了停车场的运营效率，降低了管理成本，而且为后续智能停车解决方案的开发提供了技术基础和实践经验。

关键词： 物联网；停车场监控系统；传感器； STM32F103C8T6 单片机

Design and software implementation of a parking lot monitoring data acquisition system based on the Internet of Things

Abstract

With the increasing pressure of urban traffic, the development of intelligent parking lot systems has become an important way to alleviate this problem. This article proposes a parking lot monitoring data collection system based on Internet of Things technology, aiming to improve parking efficiency and management quality by real-time monitoring of environmental information and parking space occupancy in the parking lot. The system is based on a high-performance STM32F103C8T6 microcontroller as the core control unit, integrating air temperature and humidity sensors, liquid sensors, and infrared sensors for precise monitoring of parking lot environmental parameters and parking space usage.

In summary, the IoT based parking lot monitoring data acquisition system developed in this study not only improves the operational efficiency of parking lots and reduces management costs, but also provides a technical foundation and practical experience for the development of subsequent intelligent parking solutions.

Key Words : Internet of Things; Parking lot monitoring system; Sensors; STM32F103C8T6 microcontroller

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第1章 绪论

1.1 引言

随着物联网技术的快速发展，其在智慧城市、智能交通等领域的应用日益广泛。停车场在城市交通中的作用是多方面的，它不仅仅是一个停车的地方，更是城市交通系统中的重要组成部分，对于提高城市交通效率、改善市民出行体验以及支持城市规划和商业活动都具有重要意义。因此，构建基于物联网的停车场监控数据采集系统，实现停车场环境信息和车位占用情况的实时监测与数据传输，具有重要的现实意义和应用价值。

本系统以 STM32F103C8T6 单片机（简称为 STM32 单片机）为核心控制器，结合空气温湿度传感器、液体传感器以及红外传感器等多种传感器，实现了对停车场环境温湿度、地面积水情况以及车位占用状态的全面感知^[1]。通过 OLED 屏幕实时显示采集到的数据，方便管理人员快速了解停车场当前状况。同时，利用 Message Queuing Telemetry Transport 及消息队列遥测传输协议（简称为 MQTT 协议），将通信猫云平台作为服务器，将数据转发给系统，进而实现数据的集中管理和远程监控^[2]。

通过本系统的设计与实现，不仅能够提高停车场管理的智能化水平，降低人力成本，还能够为车主提供更加便捷、安全的停车环境。同时，该系统也为类似应用场景提供了一种可行的解决方案，对于推动物联网技术在智能交通领域的应用具有一定的参考意义。

因此，本文将详细阐述基于物联网的停车场监控数据采集系统的设计与实现过程，包括系统架构设计、硬件设计与实现、软件设计与实现等方面的内容，并对系统性能进行测试与分析，以期为该领域的研究和实践提供一定的参考和借鉴。

1.2 背景与意义

随着城市化的快速推进和机动车数量的持续增长，停车问题已成为影响城市交通秩序和居民生活质量的重要因素。旧的停车场的管理方法已经难以适应现代城市管理的需求。智能化、信息化的停车管理系统成为解决这一问题的关键。物联网技术，作为新一代信息技术的核心部分，具备实时性、网络化和智能化的显著特征，为停车场监控数据采集提供了新的解决方案。物联网技术通过无线传感器网络、云计算、大数据等技术手段，能够实现对停车场内环境参数的实时监测和数据采集，以及停车位信息的智能管理和服务^[3]。该系统能够实现对停车场内环境参数和停车位信息的实时监测和数据采集。这不仅可以减少人工巡视的工作量，提高管理效率，还可以通过数据分析，为管理者提供更加准确、全面的停车场运行状况，帮助其做出更加科学、合理的管理决策。

通过对停车场内温度、湿度、积水等环境参数的实时监测，系统能够及时发现可能存在的安全隐患，如车辆锈蚀、底盘受损等。这有助于车主和管理者及时采取应对措施，避免或减少车辆损失，保障车辆停放的安全。本研究旨在通过物联网技术实现停车场监控数据采集系统的设计和实现，展示物联网技术在停车管理领域的应用效果和潜力。这有助于推动更多相关技术的研发和应用，促进物联网技术与停车管理领域的深度融合，为未来的智慧停车场建设提供技术支撑和经验借鉴。

综上所述，基于物联网的停车场监控数据采集系统的研究和应用具有重要的现实意义和应用价值。它不仅有助于提升停车场的管理效率和车辆停放的安全性，还能优化停车体验，推动物联网技术在停车管理领域的应用和发展，促进智慧城市的建设和发展。同时，本研究还具有一定的学术价值和理论意义，有助于丰富和完善物联网技术在交通管理领域的应用理论体系和实践经验。

1.3 国内外研究现状

随着国内城市化进程的加快和汽车数量的不断增长，对停车场系统的需求也在日益提升。近年来，智能化技术的运用与更新成为了国内停车场系统研究的主要焦点。智能停车诱导系统、无人值守停车场等新型停车场系统正逐步推

广应用，以提高停车场的运营效率和管理水平^[4]。此外，国内也在积极探索物联网、大数据、云计算等先进技术在停车场系统中的应用，以实现更加高效、便捷、安全的停车体验。

然而，国内的停车场系统研究仍面临一些挑战。一方面，尽管停车场系统数量在不断增加，但停车位供需矛盾依然突出，许多车主仍然面临找不到停车位的困境。另一方面，国内停车场系统的智能化水平仍有待提高，特别是在数据处理、分析和预测方面，还需要进一步加强研发和应用。

在国外，停车场系统的研究起步较早，技术相对成熟。许多国家已经建立了完善的物联网基础设施和智能停车系统，实现了对停车场环境的实时监测和车位状态的精准感知。此外，国外还注重停车场系统的集成和互联互通，将其与智能交通系统、城市管理系统等相融合，以实现更加高效的城市管理和交通疏导。

值得注意的是，国外在停车场系统的研究上更加注重用户体验和服务创新。例如，通过智能停车 APP 提供停车位预定、支付、导航等一站式服务，为车主提供更加便捷、舒适的停车体验^[5]。同时，国外还积极探索停车场系统的商业模式创新，如共享停车、车位租赁等，以缓解城市停车难问题。

综上所述，国内外在停车场系统的研究上都在积极推进智能化、集成化和服务创新等方面的发展。然而，由于国情和技术水平的差异，国内外在停车场系统的发展上还存在一定的差距。未来，我国智能停车场发展会越来越好。

1.4 本论文章节安排

本论文是在基于物联网的停车场监控数据采集系统的基础上进行编写，主要实现停车场的环境监测以及车位使用状况。本论文分为以下七个章节。

第一章 绪论：主要介绍基于物联网的停车场监控数据采集系统的背景和意义，以及国内外对于停车场智能化和服务创新方面的差异。

第二章 系统开发平台及技术：主要介绍基于物联网的停车场监控数据采集系统的开发环境、编译语言。

第三章 需求分析：主要是对系统的管理员和普通用户进行需求分析和可行性分析。

第四章 系统设计：主要是对系统进行系统设计、详细设计和数据库设计。

第五章 硬件设计与实现：主要介绍的是系统硬件功能的实现效果。

第六章 软件设计与实现：主要介绍的是系统软件功能的实现效果。

第七章 系统测试：主要介绍的是按照既定测试标准进行测试的标准和结果的展示。

第2章 系统开发平台及技术

2.1 IDEA 介绍

IDEA 软件提供了丰富的导航模式和智能的代码选取功能，使得开发人员能够更加便捷地进行工作。

IDEA 有两个版本，分别是旗舰版和社区版。这两个版本都提供了一系列核心功能，并且都支持 Git、Mercurial 和 SVN 等版本控制系统。然而，在支持的语言和框架方面，两个版本有所不同。社区版主要支持 Java、Kotlin、Groovy 和 Scala 等语言。而旗舰版，即专业版，提供了对更多语言和框架的支持，更适合多语言开发人员和全栈 Web 开发。此外，专业版还内置了数据库工具，方便与数据库交互的开发人员进行数据库查看、SQL 脚本运行、数据导出等任务。

在 IDEA 中，开发人员可以利用丰富的快捷键提高编程效率。同时，IDEA 也提供了智能导航功能，如通过 Ctrl + 鼠标左键点击可以跳转到定义，Alt + F7 可以查找引用等。此外，IDEA 还支持实时模板功能，可以通过输入缩写自动生成代码。

2.2 Keil 介绍

首先，Keil 为用户提供了完整的开发工具链，包括 C 编译器、宏汇编器、链接器、库管理器以及功能强大的仿真调试器等。这些工具完全支持 Windows 环境，为基于 Cortex-M、Cortex-R4、ARM7、ARM9 等处理器的设备开发提供了强大的支撑。

接着，Keil 软件的一大特色是其多语言支持。它不仅支持 C 语言，还支持 C++ 和汇编语言（ASM），这使得它能够满足不同编程需求。无论是刚入门的新手还是经验丰富的专业人士，都可以通过 Keil 软件对各种单片机进行编译、调试和仿真，从而完成从简单到复杂的各类项目开发。

此外，Keil 的用户友好界面也是其广受欢迎的一个重要因素。它提供了一个直观的操作环境，使得源代码编辑、程序编译以及代码调试变得更加高效和便捷。同时，Keil 还提供了丰富的 API 和库函数，方便开发者访问硬件资源，并通过模拟器和仿真器等工具来测试和验证代码的正确性，这对于提高开发效率和确保代码质量都非常有帮助。

综上所述，Keil 软件是一款强大的嵌入式系统开发工具，它以其全面的功能、高效的编译和调试能力以及用户友好的界面，成为电子相关领域学习和工程师开发工作中的重要工具。

2.3 SSM 介绍

SSM 框架是一种广泛应用于 Java Web 开发的经典技术组合，由 Spring、SpringMVC 和 MyBatis 三个开源项目整合而成。每个组件都有其独特的职责：Spring 提供了依赖注入和面向切面编程等功能，负责应用的整体配置和事务管理；SpringMVC 基于模型-视图-控制器架构，用于组织 Web 应用的代码；SSM 框架具有代码整洁简单、灵活性强、容易维护和高性能等特点，广泛应用于电商、社交、外卖等各种互联网商业场景^[6]。

2.4 Java 介绍

Java 是一门功能强大且简单易用的编程语言，具有面向对象、平台无关性、内存管理、安全性等特点。它支持自动垃圾回收机制，减轻了开发者的内存管理负担。此外，Java 的安全性机制可以防止未经授权的访问和恶意代码的执行。总的来说，Java 是一种广泛应用的编程语言，以其独特的优势在各个领域发挥着重要作用。

2.5 传感器介绍

DHT11 是一款常用的温湿度传感器，能够测量环境的温度和湿度，并将这些数据以数字信号输出，广泛应用于各种环境监测和控制系统中。

液体传感器是专门用来检测和监控液体状况的装置，它可以探测多种液体的不同参数，例如存在性、流量、温度和浓度等。在工业应用中，液体传感器

扮演着至关重要的角色，因为它能实时监控并调控液体的各种指标，从而保障生产流程的顺畅进行。

红外线反射传感器利用红外线的反射机制来探测物体或测量距离。其工作原理是，发射器发出红外光线，当这些光线触及目标物后，部分光线会反射回传感器。接收端会捕获这些反射的红外光，并将其转化为电信号。进一步处理这些电信号，传感器便能确定物体的存在或计算出与目标物的距离。这种传感器以非接触方式工作，响应迅速，探测范围广，且对环境有良好的适应性，因此在自动化控制、安全监控及机器人技术等多个领域都有广泛应用。

总的来说，dht11 传感器、液体传感器和红外传感器在各自的领域都有其独特的应用价值，在本系统中，dht11 传感器主要用于温度和湿度的测量，液体传感器用于监测积水，而红外传感器则用于监测停车位^[7]。

2.6 MQTT 服务器介绍

通信猫云平台作为一个 MQTT 服务器，具备轻量级、高可靠性和易扩展的特点。它提供稳定且有序的双向通信连接，支持多种 QoS 服务质量等级，确保数据的可靠传输。同时，通信猫云平台采用高效的协议交换和最小化的传输开销，有效降低网络流量，适应物联网设备带宽有限的特点。此外，它还支持实现一对多的消息发布，便于服务器灵活控制和管理各种设备。通信猫云平台以其高效稳定的表现，为基于物联网的停车场监控数据采集系统提供了强大的数据支持。

2.7 本章小结

本章节主要介绍本系统使用的软硬件技术的特点，对于我的这个系统，我认为这些工具和技术可以满足我的开发需求，这些工具和技术各有特色，并在本系统不同地方发挥着重要作用。

第3章 需求分析

3.1 管理员功能需求分析

本系统的用户群体包含普通用户以及管理员。接下来将进行用例分析，管理员功能用例图如图 3-1 所示。

用例分析：

（1）报警管理：管理员可对本系统温湿度阈值进行设置，超出或小于阈值则发出警报。

（2）环境历史数据查看：管理员可以查看系统环境历史数据，包括温湿度、车位积水情况，数据以表格方式呈现。

（3）环境状况实时监测：管理员可查看实时环境状况，包括温湿度、车位积水情况，监测方式为折线图，按固定时间间隔实时显示。

（4）车位和数据监测：管理员可以总览温湿度和积水情况实时数据，以及车位占用情况。

（5）用户管理：管理员可以添加、删除和修改普通用户的信息。

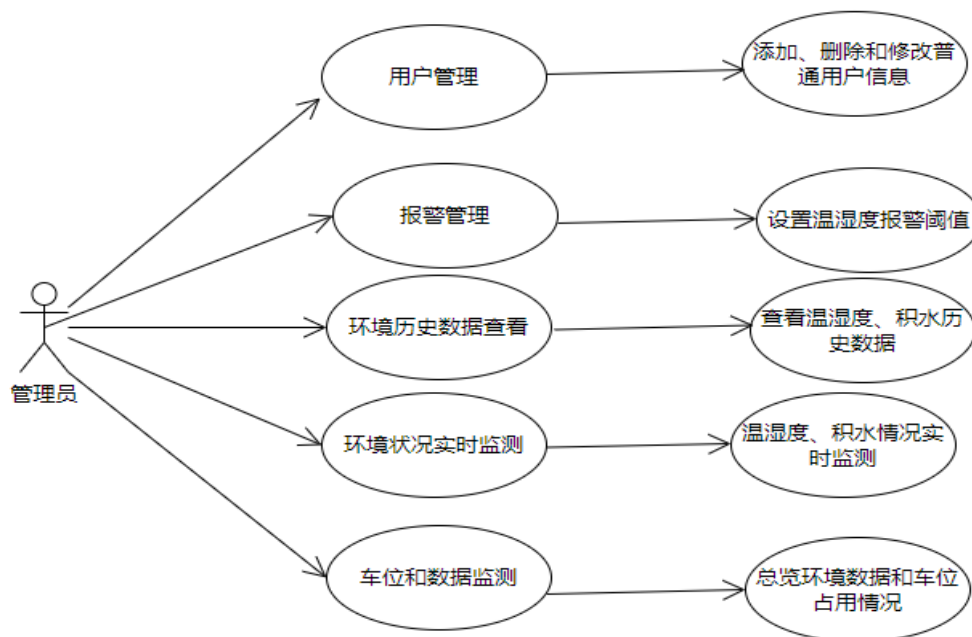


图 3-1 管理员功能用例图

3.2 普通用户功能需求分析

以下是普通用户用例分析，普通用户功能用例图如图 3-2 所示。

用例分析：

（1）报警管理：普通用户可对本系统温湿度阈值进行设置，超出或小于阈值则发出警报。

（2）环境历史数据查看：普通用户可以查看系统环境历史数据，包括温湿度、车位积水情况，数据以表格方式呈现。

（3）环境状况实时监测：普通用户可查看实时环境状况，包括温湿度、车位积水情况，监测方式为折线图，按固定时间间隔实时显示。

（4）车位和数据监测：普通用户可以总览温湿度和积水情况实时数据，以及车位占用情况。

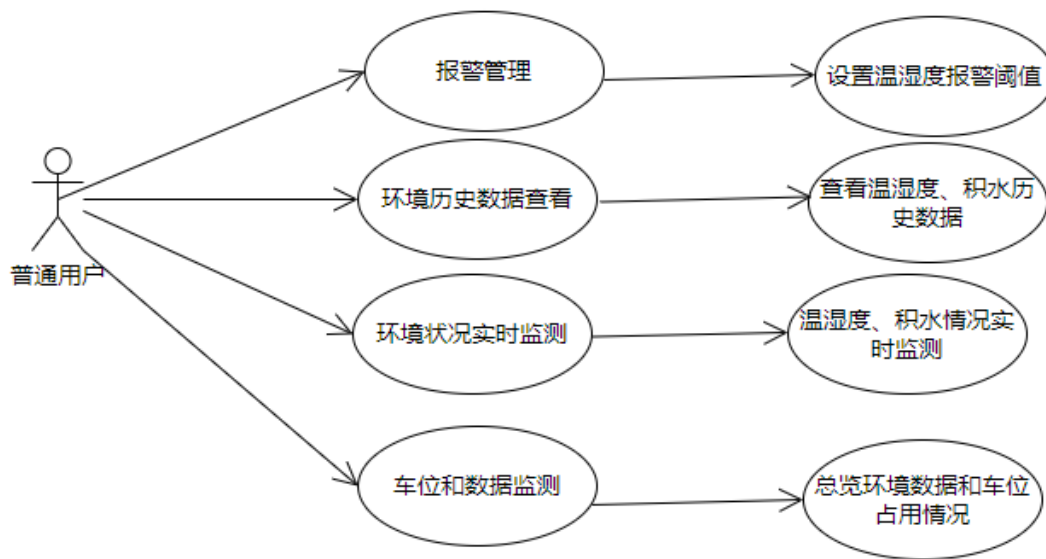


图 3-2 普通用户功能用例图

3.3 可行性分析

3.3.1 技术可行性

本系统在设计时用到了 SSM 框架、Vue、MySQL 和传感器等技术，这些技术在这几年大学学习中基本都学过，平时的编程作业都是独立思考完成，对

于一些硬件方面的知识，请教了我以前的同学，所以在技术上是可行的。通过选择合适的硬件平台、传感器技术、通信协议和软件开发环境，有信心完成该系统的设计。

3.3.2 经济可行性

本系统的投资成本主要包括 STM32 单片机、传感器、OLED 屏幕等硬件设备的购置成本。为了完成该系统，花了几十元，这些投资成本是合理的。除此之外，本系统主要需要考虑的是电能消耗、维护费用等。由于系统采用低功耗的 STM32 单片机和传感器，因此电能消耗相对较低；同时，我在放置这些硬件时比较小心，可以有效降低维护费用。

3.3.3 操作可行性

本系统选用 STM32 单片机作为主控器，并配备了空气温湿度传感器、液体传感器以及红外传感器等作为环境感知设备，以及 OLED 屏幕作为本地显示设备。这些硬件设备的部署和配置相对简单，只需按照系统设计方案进行连接和调试即可。同时，系统支持模块化设计，各模块之间相对独立，方便维护。

3.4 本章小结

本章主要阐述了对整个系统进行的需求分析和可行性分析，首先明确本系统的使用用户为管理员和普通用户，分别介绍一下大体需求，并使用 UML 用例图更直观地表现出来。之后我从技术可行性、经济可行性和操作可行性三个方面详细分析，这样的话能确保本系统能够成功实施并达到预期目标。

第4章 系统设计

4.1 系统总体设计

基于物联网的停车场监控数据采集系统的用户有普通用户和管理员，普通用户可以进行报警管理、环境历史数据查看、环境实时状况监测以及车位和数据监测，管理员可以进行用户管理、报警管理、环境历史数据查看、环境实时状况监测以及车位和数据监测^[8]。具体如图 4-1 所示。

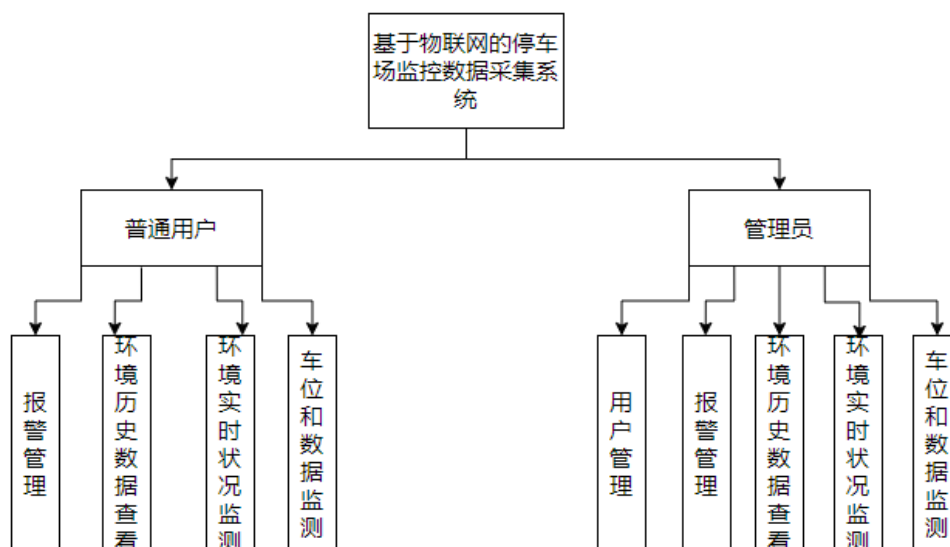


图 4-1 停车场监控数据采集系统功能架构图

为了完成以上这些功能，设计切入点为停车场温度、湿度、是否积水、停车位信息以及报警管理。系统对停车场内温湿度和积水情况进行检测，通过空气温湿度 dht11 传感器和液体传感器来收集数据，并且系统内部允许设定温湿度的阈值。超过阈值蜂鸣器会报警，3 个红外传感器模拟检测 3 个车位是否停车，OLED 屏幕显示器会显示车位占用情况。使用通信猫云平台，将所有数据转发给系统^[9]。具体设计如图 4-2 所示。

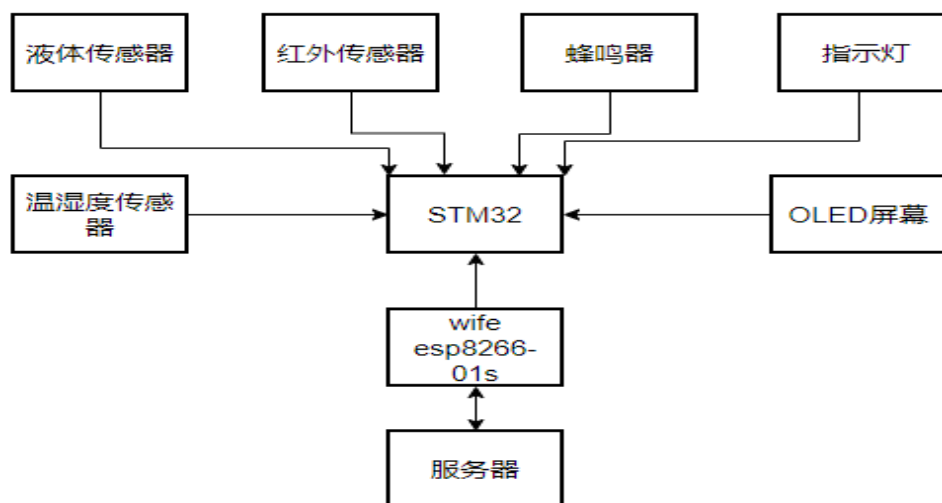


图 4-2 硬件系统功能架构图

4.2 系统详细设计

4.2.1 用户管理流程设计

管理员登录系统后，会进入用户管理界面。在这个界面中，管理员能执行多种操作，例如查看用户信息、添加新用户、更新已有用户的信息，或是删除用户。每一步操作都精确而有序，确保了用户管理的准确性和高效性。在进行查询操作时，为了确认或修改相关信息，管理员需要输入特定的用户名来进行查询。管理员可以填写或更新用户信息，然后可以保存所作的更改。整个流程设计得既清晰又直观，使得用户能够轻松理解并按照步骤进行操作，有效地提升管理的便捷性^[10]。具体设计流程如图 4-3 所示。

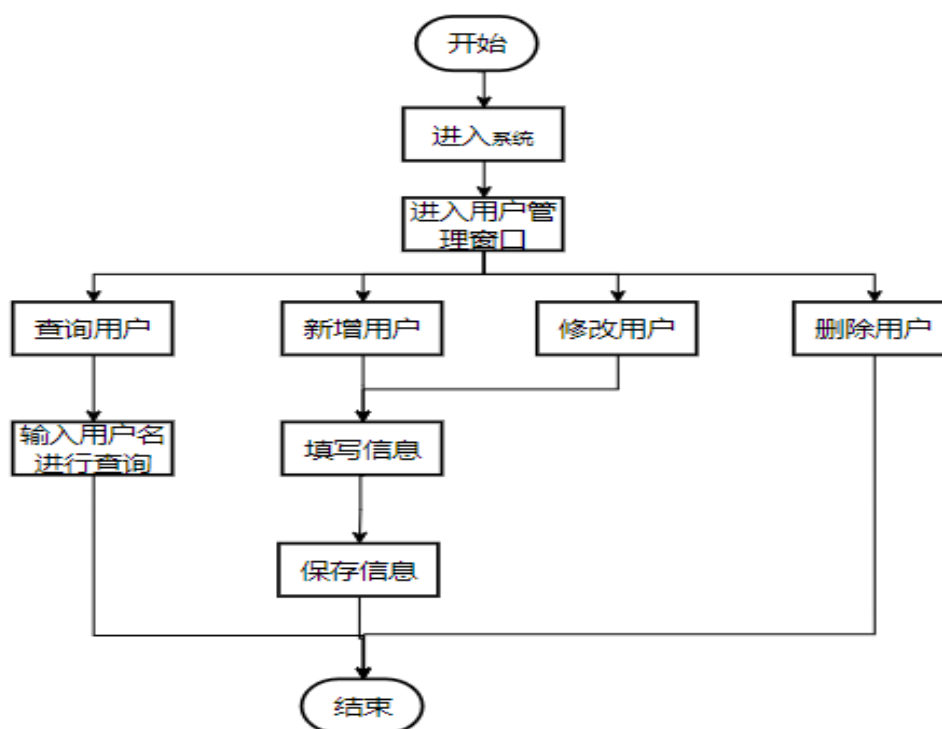


图 4-3 用户管理流程设计图

4.2.2 报警管理流程设计

用户进入系统后，进入报警管理窗口，在这个环节，用户需要勾选温度或湿度选项，然后选择进行管理。用户需要填写阈值信息，设定温度或湿度的警戒值。完成这些步骤后，用户需要保存所设置的管理参数。整个流程清晰明了，便于用户理解和操作。具体设计流程如图 4-4 所示。



图 4-4 报警管理流程设计图

4.2.3 环境历史数据查看流程设计

用户首先进入系统。接着，一个关键步骤是连接硬件设备，确保设备能够正常工作并与系统连接。用户将分别进入“温度数据窗口”、“湿度数据窗口”和“积水数据窗口”，以便在各自的窗口中查看收集到的数据。在获取到数据后，用户需要执行“刷新数据”的操作，以确保获取的是最新的环境信息。以上操作有助于用户高效地完成环境数据的收集和处理工作。通过这一流程，用户可以方便地查看温度、湿度和积水等关键环境指标，从而及时做出响应和调整。具体设计流程如图 4-5 所示。

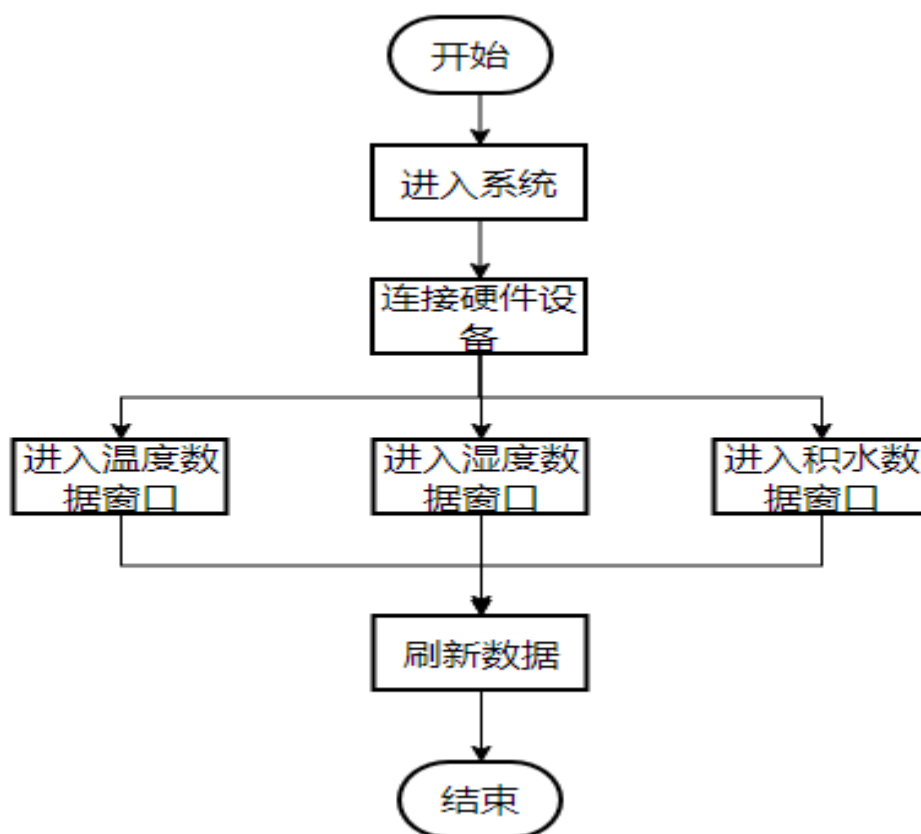


图 4-5 流程设计图

4.2.4 环境实时状况监测流程设计

用户首先进入系统。接着，一个关键步骤是连接硬件设备，确保设备能够正常工作并与系统连接。之后，程序分别进入“温度监控窗口”、“湿度监控窗口”和“积水监控窗口”，对环境参数进行实时监测。在监控过程中，程序会对各项指标进行评估，检查它们是否在正常范围内。一旦发现某项指标异常，即超出正常范围，系统就会触发警报。此外，系统会实时绘制折线图，并将当前监测到的数值展示给用户，方便用户随时了解环境状况。具体设计流程如图 4-6 所示。

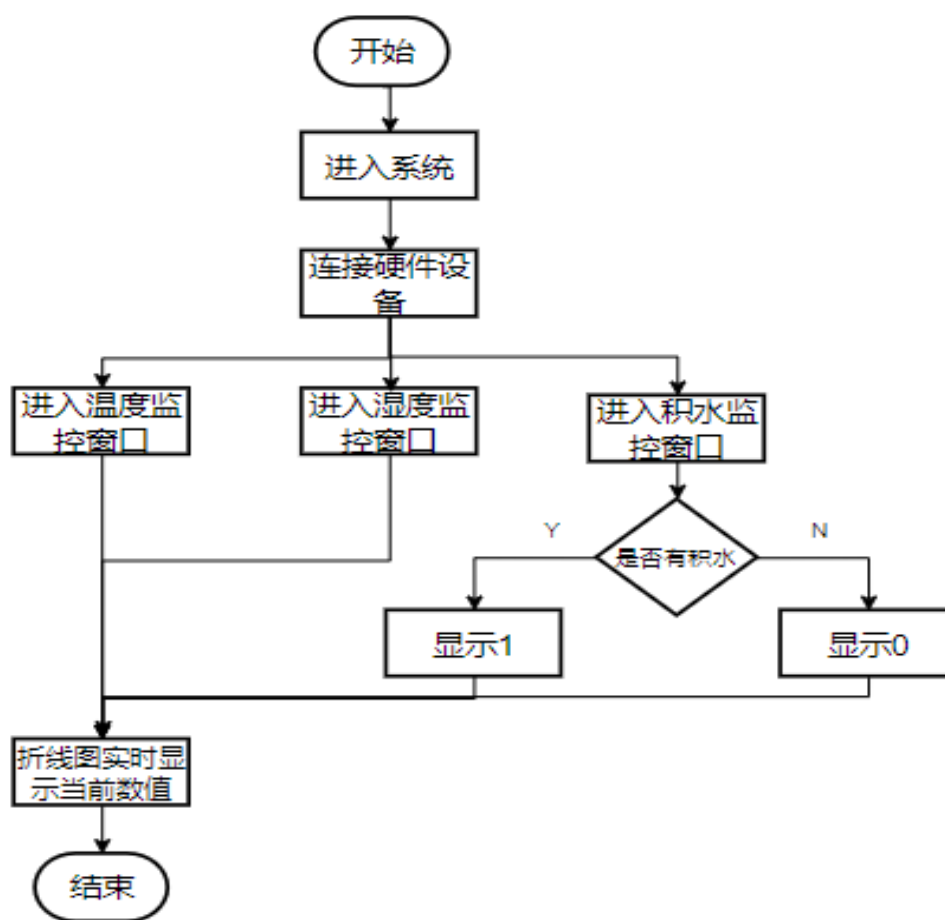


图 4-6 环境实时状况监测流程设计图

4.2.5 车位和数据监测流程设计

首先，用户需要连接硬件设备并进入系统。接着，进入车位和数据监控界面，通过监控窗口查看车位状况。在此过程中，用户还可以获取所有环境实时数据。接下来，根据车位是否被占用的情况，系统会显示不同结果。如果车位未被占用，系统会显示无车，表示该车位可用；如果车位已被占用，则会显示有车，表示该车位不可用^[11~13]。具体设计流程如图 4-7 所示。

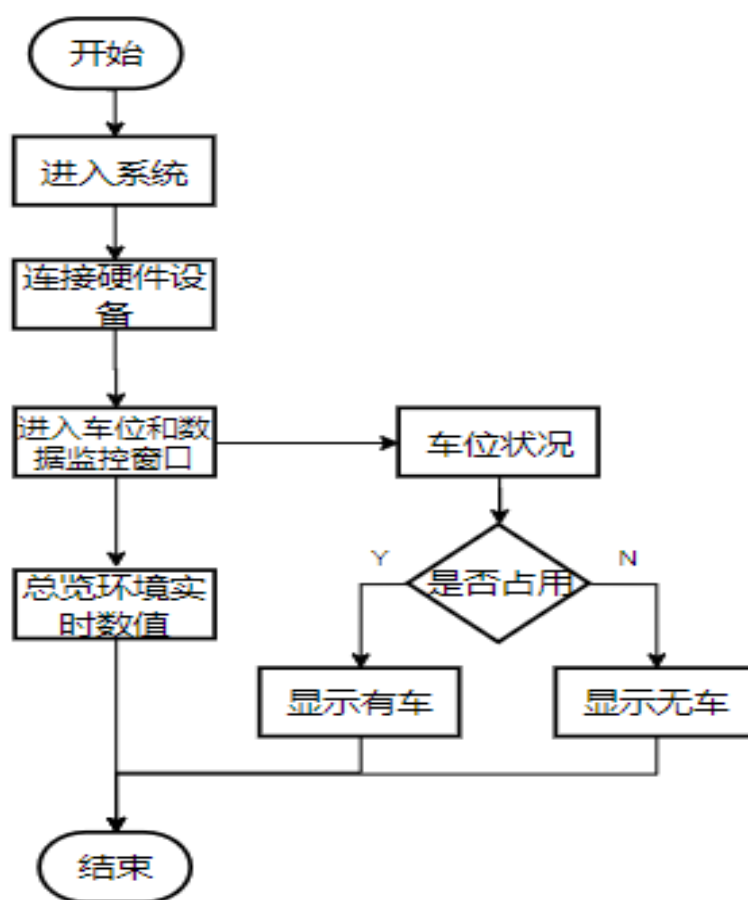


图 4-7 车位和数据监测流程设计图

4.3 数据库设计

4.3.1 数据库概念设计

（1）实体和属性的定义

用户角色关系表（序号，用户 ID，角色 ID）

用户表（用户 ID，用户名，密码，邮箱，手机号，状态，创建者 ID，创建时间）

角色菜单关系表（序号，角色 ID，带单 ID）

角色表（角色 ID，角色名，备注，创建者 ID，创建时间）

菜单表（菜单 ID，父菜单 ID，菜单名，菜单地址，授权，类型，菜单图标，排序）

配置表（序号，key 值，key 对应值，状态，备注）

监控数据表（序号，数据值，类型，采集时间）

（2）ER 图

图 4-8 展示了用户、菜单和角色之间的关系，采用的是实体关系（ER）图。在这个图中，用户、菜单和角色都被视为实体类对象。一个角色可以涵盖多个用户，同时，角色与菜单之间存在着多对多的对应关系。

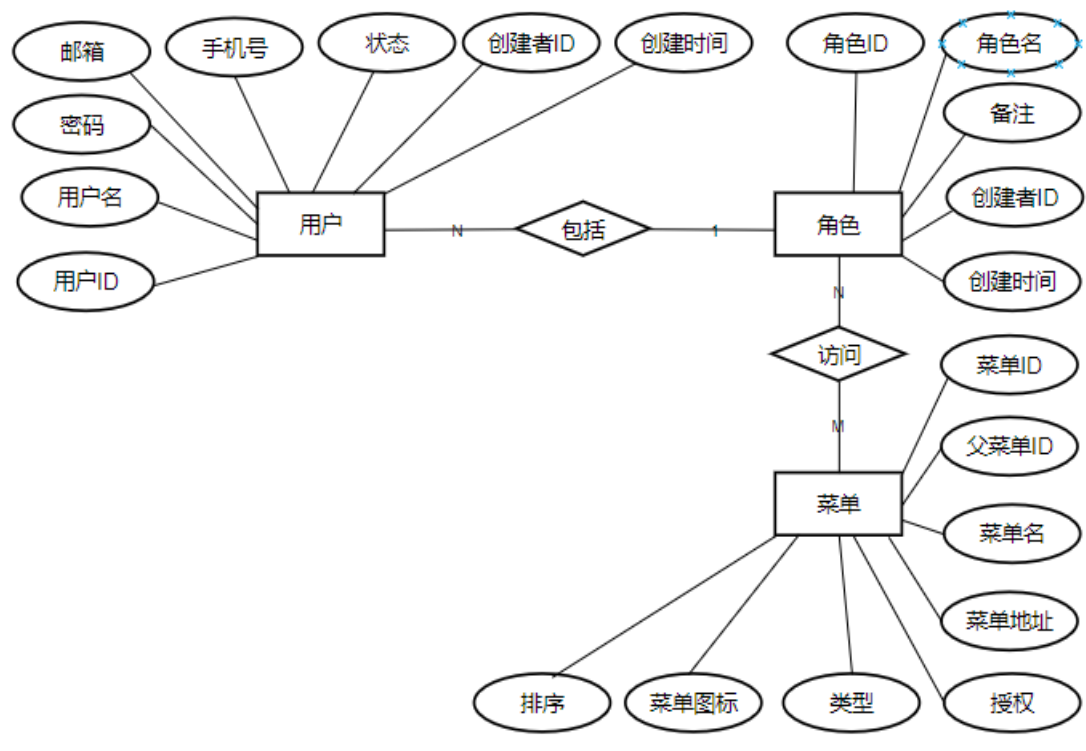


图 4-8 ER 图

4.3.2 数据库的实现

基于物联网的停车场监控数控采集系统数据库由七张表构成。下面介绍各个数据表的详细信息。

用户角色关系表（sys_user_role）记录用户与角色对应关系，表结构如表 4-1 所示。

表 4-1 用户角色关系表

字段名称	类型	字段说明	主键	是否允许为空
id	bigint		是	否
user_id	bigint	用户 ID		是
role_id	bigint	角色 ID		是

用户表（sys_user）记录用户信息，表结构如表 4-2 所示。

表 4-2 用户表

字段名称	类型	字段说明	主键	是否允许为空
user_id	bigint	用户 ID	是	否
username	varchar	用户名		否
password	varchar	密码		是
email	varchar	邮箱		是
mobile	varchar	手机号		是
status	tinyint	状态		是
create_user_id	bigint	创建者 ID		是
create_time	datetime	创建时间		是

角色菜单关系表（sys_role_menu）记录角色与菜单对应关系，表结构如表 4-3 所示。

表 4-3 角色菜单关系表

字段名称	类型	字段说明	主键	是否允许为空
id	bigint		是	否
role_id	bigint	角色 ID		是
menu_id	bigint	菜单 ID		是

角色表（sys_role）记录角色信息，表结构如表 4-4 所示。

表 4-4 角色表

字段名称	类型	字段说明	主键	是否允许为空
role_id	bigint	角色 ID	是	否
role_name	varchar	角色名		是
remark	varchar	备注		是
create_user_id	bigint	创建者 ID		是

create_time	datetime	创建时间		是
-------------	----------	------	--	---

菜单表（sys_menu）记录菜单信息，表结构如表 4-5 所示。

表 4-5 菜单表

字段名称	类型	字段说明	主键	是否允许为空
menu_id	bigint	菜单 ID	是	否
parent_id	bigint	父菜单 ID		是
name	varchar	菜单名		是
url	varchar	菜单地址		是
perms	varchar	授权		是
type	Int	类型		是
icon	varchar	菜单图标		是
order_num	int	排序		是

配置表（sys_menu）记录系统配置信息表，表结构如表 4-6 所示。

表 4-6 配置表

字段名称	类型	字段说明	主键	是否允许为空
id	bigint		是	否
key	varchar	key		是
value	varchar	value		是
status	tinyint	状态		是
remark	varchar	备注		是
min	varchar			是

监控数据表（sys_menu）记录监控数据，表结构如表 4-7 所示。

表 4-7 监控数据表

字段名称	类型	字段说明	主键	是否允许为空
id	bigint		是	否

v	varchar	数据值		是
type	varchar	类型		是
time	datetime	采集时间		是

4.4 本章小结

本章主要阐述了系统的整体设计思路、详细设计方案以及数据库的实现过程。首先，我们通过绘制系统的总体架构图来深入分析其结构。然后，画流程图来对具体功能详细设计，最后，设计出数据库的几张表和 ER 图。

第5章 硬件设计与实现

5.1 蜂鸣器

用户进入系统后设置温湿度的阈值，接下来，系统利用 STM32 单片机从空气温湿度传感器中收集数据，并将这些数据与预先设定的温湿度临界值进行对比。如果发现温湿度超出了设定的范围，STM32 单片机就会立刻启动蜂鸣器报警装置，使蜂鸣器发出连续的警报声，此外，如果液体传感器上有水，系统内也会做出提示，蜂鸣器也会发出警报声。硬件部分如图 5-1 所示。



图 5-1 蜂鸣器

```
/*  
*****  
蜂鸣器初始化  
*****/  
GPIO_InitStructure.GPIO_Pin = GPIO_Pin_5;  
GPIO_Init(GPIOB, &GPIO_InitStructure);  
BEEP=1;
```

图 5-2 蜂鸣器代码

5.2 温湿度传感器

在基于物联网的停车场监控数据采集系统中，空气温湿度传感器连接到 STM32 单片机上，利用 I/O 接口来传输数据。STM32 单片机将定时从传感器中获取数据，然后对这些数据进行所需的处理和分析。处理后的数据可以通过 OLED 屏幕进行本地

显示，也可以通过通信猫云平台上传至系统进行远程监控和管理。硬件部分如图 5-3 所示。

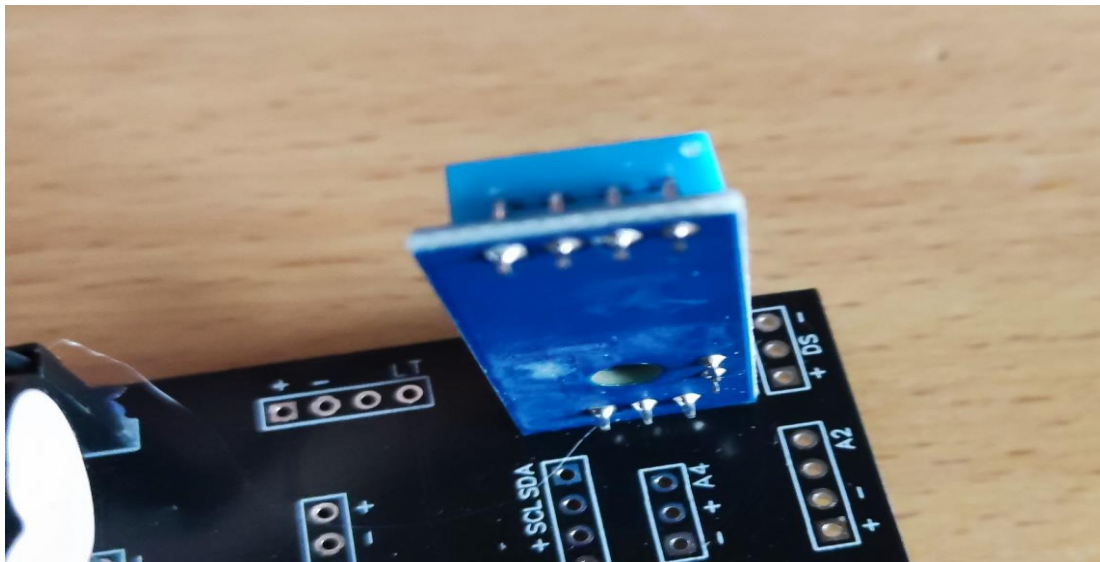


图 5-3 温湿度传感器

```
//从DHT11读取一个字节
//返回值：读到的数据
u8 DHT11_Read_Byte(void)
{
    u8 i,dat;
    dat=0;
    for (i=0;i<8;i++)
    {
        dat<<=1;
        dat|=DHT11_Read_Bit();
    }
    return dat;
}
```

图 5-4 温湿度传感器代码

5.3 液体传感器

在基于物联网的停车场监控数据采集系统中，液体传感器与 STM32 单片机相连，通过 I/O 接口实现数据的传输和交互。单片机定时读取液体传感器的状态信息，并根据预设的阈值进行判断和处理。当检测到积水时，单片机会触发警报机制。处理后的数据可以通过 OLED 屏幕进行本地显示，也可以通过通信猫云平台上传至系统进行远程监控和管理。硬件部分如图 5-5 所示。

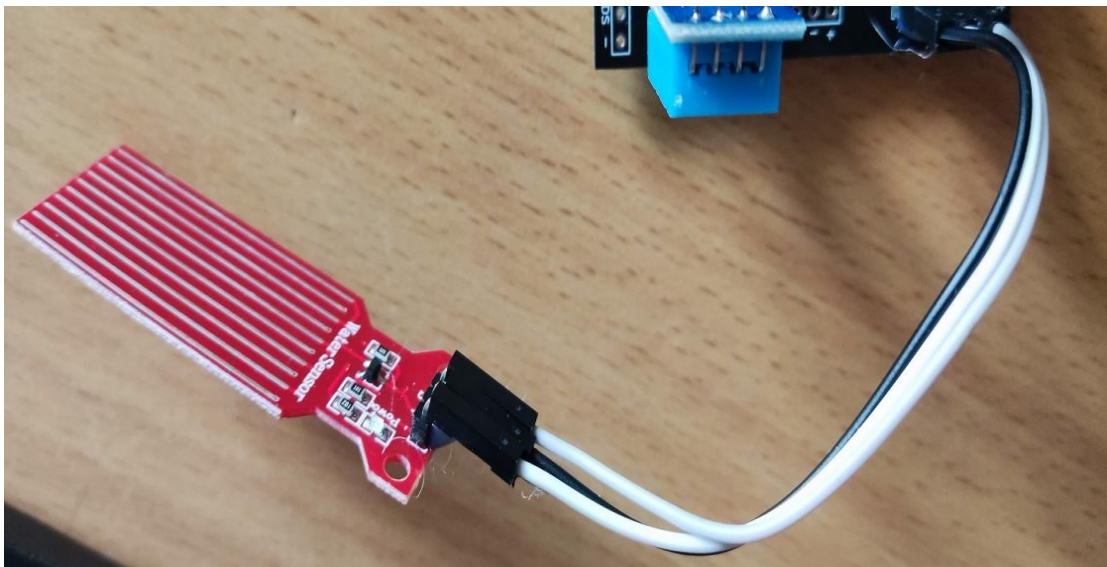


图 5-5 液体传感器

```
void INPUT_Init(void)
{
    GPIO_InitTypeDef GPIO_InitStructure;

    RCC_APB2PeriphClockCmd(RCC_APB2Periph_GPIOB|RCC_APB2Periph_GPIOA,ENABLE);

    GPIO_PinRemapConfig(GPIO_Remap_SWJ_JTAGDisable, ENABLE);

    GPIO_InitStructure.GPIO_Pin = GPIO_Pin_10|GPIO_Pin_11|GPIO_Pin_15;//PB10 PB11 PB15
    GPIO_InitStructure.GPIO_Mode = GPIO_Mode_IPU;
    GPIO_Init(GPIOB, &GPIO_InitStructure);//初始化GPIOA15
```

图 5-6 液体传感器代码

5.4 红外传感器

红外传感器安装在合适位置，确保能够准确检测到车辆的进入和离开。当判断存在障碍物时，红外传感器会检测到车辆的存在，并将相应的信号传输给 STM32 单片机进行处理。在基于物联网的停车场监控数据采集系统中，红外传感器与 STM32 单片机通过 I/O 接口进行连接，实现数据的传输和交互。单片机定时读取红外传感器的状态信息，并根据读取的结果判断车位是否被占用。当红外传感器检测到车位被占用时，单片机将相应的状态信息通过 OLED 屏幕进行本地显示，方便查看占用情况，也可以通过通信猫云平台上传至系统进行远程监控和管理。硬件部分如图 5-7 所示。

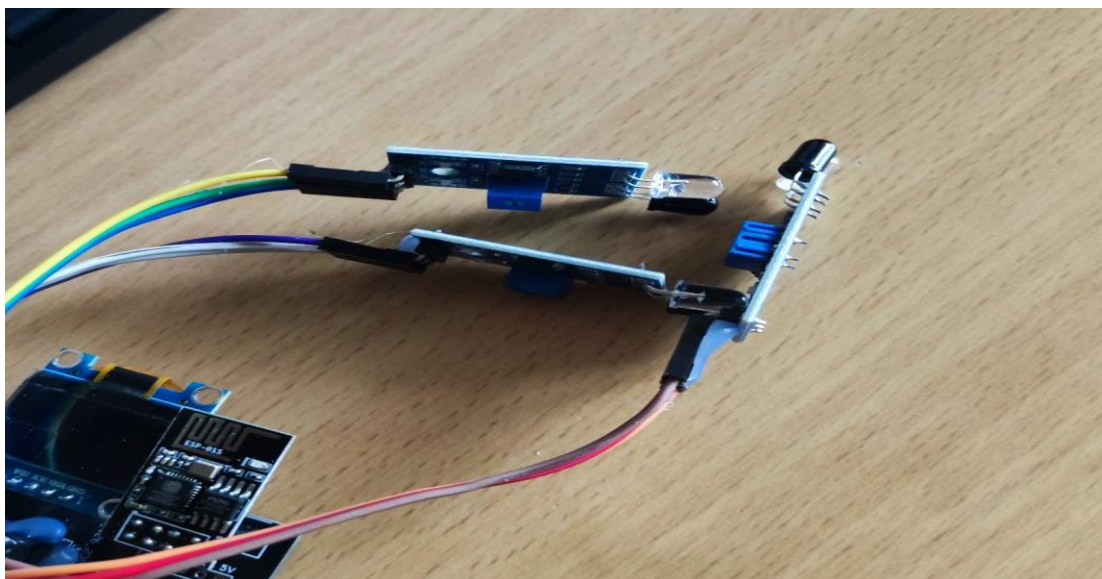


图 5-7 红外传感器

```
void INPUT_Init(void)
{
    GPIO_InitTypeDef GPIO_InitStructure;

    RCC_APB2PeriphClockCmd(RCC_APB2Periph_GPIOB|RCC_APB2Periph_GPIOA,ENABLE);

    GPIO_PinRemapConfig(GPIO_Remap_SWJ_JTAGDisable, ENABLE);

    GPIO_InitStructure.GPIO_Pin = GPIO_Pin_10|GPIO_Pin_11|GPIO_Pin_15;//PB10 PB11 PB15
    GPIO_InitStructure.GPIO_Mode = GPIO_Mode_IPU;
    GPIO_Init(GPIOB, &GPIO_InitStructure);//初始化GPIOA15
```

图 5-8 红外传感器代码

5.5 指示灯

指示灯在本基于物联网的停车场监控数据采集系统中发挥着直观且重要的指示作用。当硬件成功连接后，蓝色指示灯将以每秒一次的频率闪烁，这不仅表示系统正在正常运行，同时也明确指示数据采集正在进行中。硬件部分如图 5-9 所示。

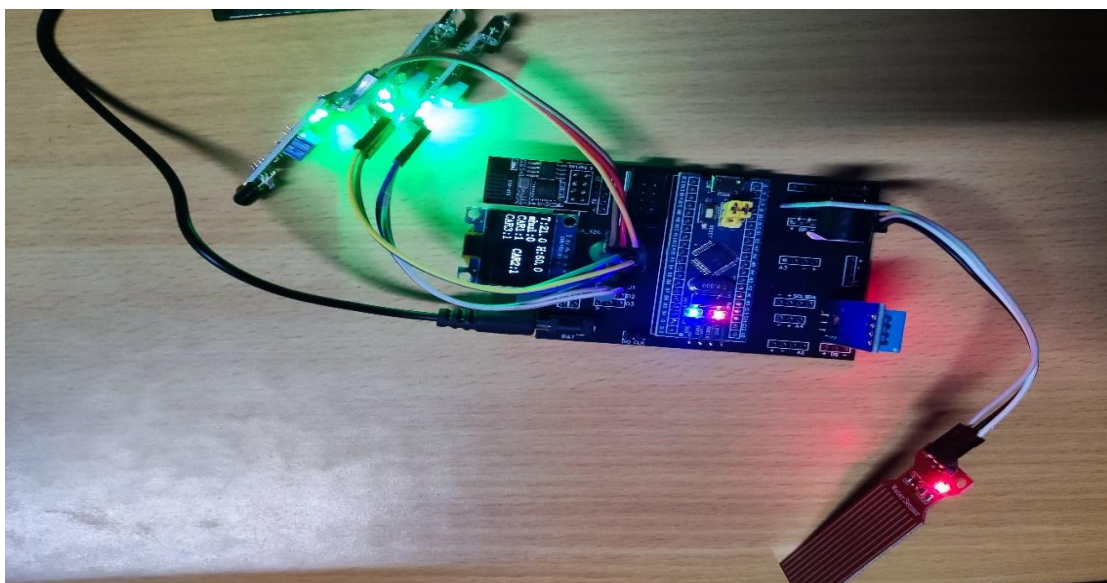


图 5-9 指示灯

```
//LED闪烁任务及MQTT心跳处理函数
//当通讯正常，则单片机信号灯会1秒1闪
//每秒发送一次数据给上位机
void LED_task(void *pvParameters)
{
    // char task_str[400];
    static u8 Counter_MQTT_Heart=0;
    while(1)
    {
        LED=!LED;
        Publish_All();

        Counter_MQTT_Heart++;
        if(Counter_MQTT_Heart>=200)
        {
            if(wifi_)
                MQTT_SentHeart();
            Counter_MQTT_Heart=0;
        }
        delay_ms(1000);
    }
}
```

图 5-10 指示灯代码

5.6 MQTT 服务器

在本基于物联网的停车场监控数据采集系统中，虽然 MQTT 服务器在视觉上无法直接感受，但是它发挥着核心的数据中转作用。通信猫平台接收来自 STM32 单片机上传的停车场环境信息和车位占用状态数据，包括空气温湿度、积水情况以及车位

状态等^[14]。随后，通信猫平台作为服务器将这些数据实时转发给系统。服务器如图 5-11 所示。



图 5-11 MQTT 服务器

```
//MQTT初始化
void MQTT_Init(uint8_t *prx,uint16_t rxlen,uint8_t *ptx,uint16_t txlen)
{
    memset(Usart2_TxBuff,0,sizeof(Usart2_TxBuff));
    memset(Usart2_RxBuff,0,sizeof(Usart2_RxBuff));

    MQTT_Disconnect();
    //HAL_Delay(100);
    vTaskDelay(100);
    MQTT_Disconnect();
    //HAL_Delay(100);
    vTaskDelay(100);
}
```

图 5-12 MQTT 服务器代码

5.7 Wife

在本基于物联网的停车场监控数据采集系统中，我使用的网络为 Wife sep8266-01s^[15~16]。它作为系统的主要通信方式，实现了 STM32 单片机与通信猫云平台之间的连接，确保空气温湿度、积水检测以及车位占用状态等实时数据能够高效、稳定地上传至系统。Wife 模块如图 5-13 所示。

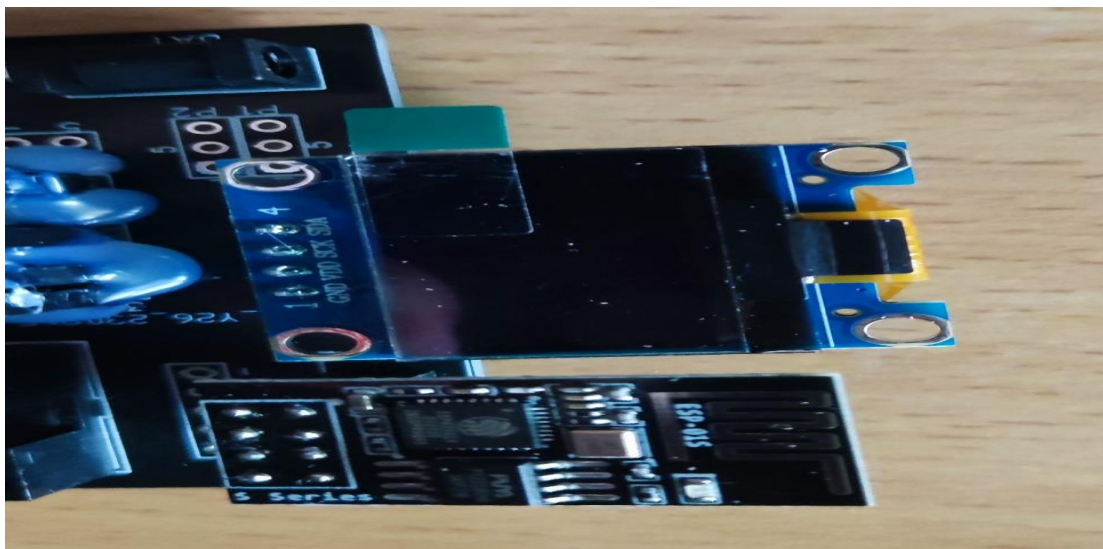


图 5-13 Wife 模块图

```
char WiFi_Get_Data(char *data, char *len, char *id)
{
    char temp[10]={0};    //缓冲区
    char *presult;

    if(strstr(U2_RX_BUF,"\r\n\r\n")){ //两个连着的回车换行作为数据的结束符
        sscanf(U2_RX_BUF,"%[^,],%[^,],%[^:]",temp,id,len); //截取各段数据，主要是id和数据长度
        presult = strstr(U2_RX_BUF,":"); //查找冒号。冒号后的是数据
        if( presult != NULL ) //找到冒号
            sprintf((char *)data,"%s", (presult+1)); //冒号后的数据，复制到data
        U2_RxCounter=0; //WiFi接收数据量变量清零
        memset(U2_RX_BUF,0,U2_RXBUFF_SIZE); //清空WiFi接收缓冲区
        return 1; //有数据到来
    } else return 0; //无数据到来
}
//
```

图 5-14 Wife 模块代码

5.8 本章小结

本章详细描述了基于物联网的停车场监控数据采集系统的硬件设计与实现过程。首先，我们选择了 STM32 单片机作为系统的核心控制器，接着，我们选用了适当的传感器来检测停车场的环境信息和车位占用情况。空气温湿度传感器能够实时采集停车场内的温度和湿度数据，液体传感器则用于检测停车场内是否有积水，红外传感器则用于检测车位的占用情况。同时，我们还利用 OLED 屏幕实时显示采集到的数据。此外，为了实现远程数据传输，我们采用了 MQTT 协议作为通信协议，并通过通信猫云平台将数据上传至系统。

第6章 软件设计与实现

6.1 用户管理

在用户管理界面上方，有“查询”“新增”“修改”“删除”按钮，下方的表格则清晰地展示了用户信息，包括用户名和邮箱等字段。在用户管理界面管理员可以输入用户名查询某一个用户，点新增按钮增加用户，输入用户信息进行保存，点击用户名前一列的选择框，再点击修改和删除按钮进行修改和删除用户信息^[17]。用户管理界面如图 6-1 所示。

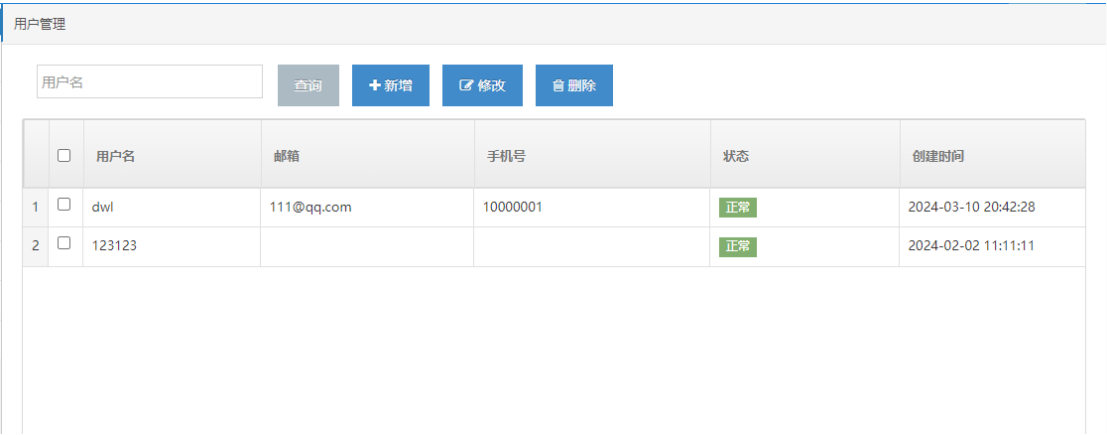


图 6-1 用户管理界面

```
/**
 * 用户信息
 */
@RequestMapping("/info/{userId}")
public R info(@PathVariable("userId") Long userId) {
    SysUserEntity user = sysUserService.queryObject(userId);

    // 获取用户所属的角色列表
    List<Long> roleIdList = sysUserRoleService.queryRoleIdList(userId);
    user.setRoleIdList(roleIdList);

    return R.ok().put("user", user);
}
```

图 6-2 用户信息查看代码

6.2 报警配置

在报警配置界面，有“修改”按钮，下方的表格则清晰地展示了温湿度阈值信息，在报警配置界面用户可以点击参数名前一列的选择框，再点击修改按钮进行修改阈值信息。报警配置界面如图 6-3 所示。



图 6-3 报警配置界面

```
/**
 * 保存配置
 */
@RequestMapping("/save")
public R save(@RequestBody SysConfigEntity config){

    sysConfigService.save(config);

    return R.ok();
}
```

图 6-4 保存阈值代码

6.3 温度数据管理

在温度数据界面，有 “刷新数据” 按钮，下方的表格则清晰地展示了某一参数名和采集时间，在该界面用户可以点击刷新数据按钮，查看当前最新的温度数据。在这里，我使用的是 jqGrid 表格插件，使得能够生成表格。温度数据界面如图 6-5 所示。

温度数据

刷新数据

	☐	温度	采集时间
1	☐	20.1	2024-03-02 15:21:57
2	☐	20.1	2024-03-02 15:21:56
3	☐	20.1	2024-03-02 15:21:54
4	☐	20.1	2024-03-02 15:21:53
5	☐	20.1	2024-03-02 15:21:52
6	☐	20.1	2024-03-02 15:21:50
7	☐	20.1	2024-03-02 15:21:49
8	☐	20.1	2024-03-02 15:21:48
9	☐	20.1	2024-03-02 15:21:47
10	☐	20.1	2024-03-02 15:21:45

图 6-5 温度数据界面

```
/**
 * 所有配置列表
 */
@RequestMapping("/list")
public R list(@RequestParam Map<String, Object> params){
    // 查询列表数据
    Query query = new Query(params);
    List<SysConfigEntity> configList = sysConfigService.queryList(query);
    int total = sysConfigService.queryTotal(query);

    PageUtils pageUtil = new PageUtils(configList, total, query.getLimit(), query.getPage());

    return R.ok().put("page", pageUtil);
}
```

图 6-6 温度历史数据代码

6.4 湿度数据管理

在湿度数据界面，有 “刷新数据” 按钮，下方的表格则清晰地展示了某一参数名和采集时间，在该界面用户可以点击刷新数据按钮，查看当前最新的湿度数据。在这里，我使用的是 jqGrid 表格插件，使得能够生成表格。湿度数据界面如图 6-7 所示。

湿度数据

刷新数据

	☐	湿度	采集时间
1	☐	52.0	2024-04-01 21:12:21
2	☐	52.0	2024-04-01 21:12:20
3	☐	52.0	2024-04-01 21:12:19
4	☐	52.0	2024-04-01 21:12:17
5	☐	52.0	2024-04-01 21:12:16
6	☐	52.0	2024-04-01 21:12:15
7	☐	51.0	2024-04-01 21:12:14
8	☐	51.0	2024-04-01 21:12:12
9	☐	51.0	2024-04-01 21:12:11
10	☐	51.0	2024-04-01 21:12:10

图 6-7 湿度数据界面

```
/**
 * 所有配置列表
 */
@RequestMapping("/list")
public R list(@RequestParam Map<String, Object> params){
    // 查询列表数据
    Query query = new Query(params);
    List<SysConfigEntity> configList = sysConfigService.queryList(query);
    int total = sysConfigService.queryTotal(query);

    PageUtils pageUtil = new PageUtils(configList, total, query.getLimit(), query.getPage());

    return R.ok().put("page", pageUtil);
}
```

图 6-8 湿度历史数据代码

6.5 积水数据管理

在积水数据界面，有 “刷新数据” 按钮，下方的表格则清晰地展示了是否积水和采集时间，在该界面用户可以点击刷新数据按钮，查看当前最新的积水数据。在这里，我使用的是 jqGrid 表格插件，使得能够生成表格。积水数据界面如图 6-9 所示。

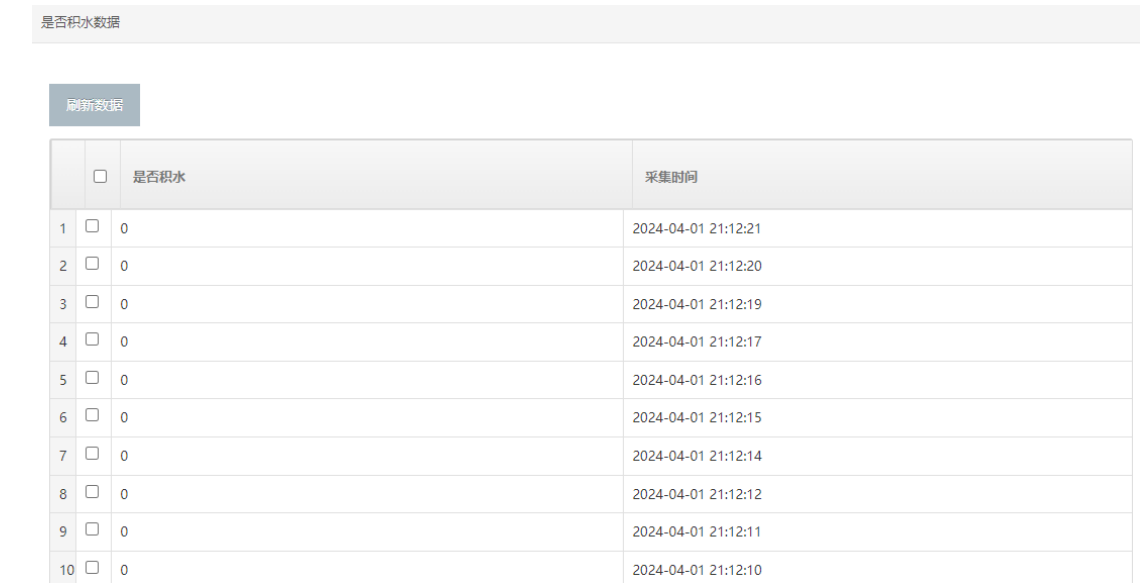


图 6-9 积水数据界面

```
/**
 * 所有配置列表
 */
@RequestMapping("/list")
public R list(@RequestParam Map<String, Object> params){
    // 查询列表数据
    Query query = new Query(params);
    List<SysConfigEntity> configList = sysConfigService.queryList(query);
    int total = sysConfigService.queryTotal(query);

    PageUtils pageUtil = new PageUtils(configList, total, query.getLimit(), query.getPage());

    return R.ok().put("page", pageUtil);
}
```

图 6-10 积水历史数据代码

6.6 温度监控

在温度监控界面，下方的曲线图则清晰地展示了当前最新的温度数据。在这里，我首先获取到最新的 10 条数据，并显示出来，当硬件正常连接后，数据开始实时变化，使用 echarts 曲线图插件，隔一秒钟显示一次^[18]。温度监控界面如图 6-11 所示。



图 6-11 温度监控界面

```
/**
 * 列表 环境历史数据
 */
@RequestMapping("/list")
public R list(@RequestParam Map<String, Object> params) {
    // 查询列表数据,
    Query query = new Query(params);

    List<DataEntity> dataList = dataService.queryList(query);
    int total = dataService.queryTotal(query);
    PageUtils pageUtil = new PageUtils(dataList, total, query.getLimit(), query.getPage());

    return R.ok().put("page", pageUtil);
}
```

图 6-12 温度实时数据代码

6.7 湿度监控

在湿度监控界面，下方的曲线图则清晰地展示了当前最新的湿度数据。在这里，我首先获取到最新的 10 条数据，并显示出来，当硬件正常连接后，数据开始实时变化，使用 echarts 曲线图插件，隔一秒钟显示一次。湿度监控界面如图 6-13 所示。



图 6-13 湿度监控界面

```
/**
 * 列表 环境历史数据
 */
@RequestMapping("/list")
public R list(@RequestParam Map<String, Object> params) {
    // 查询列表数据,
    Query query = new Query(params);

    List<DataEntity> dataList = dataService.queryList(query);
    int total = dataService.queryTotal(query);
    PageUtils pageUtil = new PageUtils(dataList, total, query.getLimit(), query.getPage());

    return R.ok().put("page", pageUtil);
}
```

图 6-14 湿度实时数据代码

6.8 积水监控

在积水监控界面，下方的曲线图则清晰地展示了当前最新的积水数据。在这里，我首先获取到最新的 10 条数据，并显示出来，当硬件正常连接后，数据开始实时变化，使用 echarts 曲线图插件，隔一秒钟显示一次。积水监控界面如图 6-15 所示。



图 6-15 积水监控界面

```
/**
 * 列表 环境历史数据
 */
@RequestMapping("/list")
public R list(@RequestParam Map<String, Object> params) {
    // 查询列表数据,
    Query query = new Query(params);

    List<DataEntity> dataList = dataService.queryList(query);
    int total = dataService.queryTotal(query);
    PageUtils pageUtil = new PageUtils(dataList, total, query.getLimit(), query.getPage());

    return R.ok().put("page", pageUtil);
}
```

图 6-16 积水实时数据代码

6.9 车位和数据监控

该界面提供了数据总览的功能，能够实时展现多个车位的状态信息。界面上清晰列出了停车场的详细数据，包括温度、湿度以及是否积水的情况。界面还通过“有车”和“无车”的标识，直观地显示三个车位当前占用情况^[19~20]。车位数据监控界面如图 6-17 所示。



图 6-17 车位数据监控界面

```
/**
 * 信息
 */
@RequestMapping("/info/{id}")
public R info(@PathVariable("id") Long id) {
    DataEntity data = dataService.queryObject(id);
    return R.ok().put("data", data);
}
```

图 6-18 车位和总体数据代码

6.10 本章小结

本章主要是对系统所设定的功能进行具体的描述和实现，包括：用户管理、报警配置、温度数据管理、湿度数据管理、积水数据管理、温度监控、湿度监控、积水监控、车位和数据监控，然后对这些功能进行部分重要代码的展示。

第7章 系统测试

7.1 用户管理功能测试

本部分主要测试的内容是用户管理功能的测试标准和测试结果。

表 7-1 用户管理功能测试表

要测试的功能	预期结果	实际结果
新增用户	√	√
修改用户	√	√
删除用户	√	√
输入用户名查询	√	√

7.2 报警配置功能测试

本部分主要测试的内容是报警配置功能的测试标准和测试结果。

表 7-2 报警配置功能测试表

要测试的功能	预期结果	实际结果
对温度的阈值进行修改	√	√
对湿度的阈值进行修改	√	√

7.3 环境数据管理功能测试

本部分主要测试的内容是环境数据管理功能的测试标准和测试结果。

表 7-3 环境数据管理功能测试表

要测试的功能	预期结果	实际结果
历史温度数据正常显示	√	√
历史湿度数据正常显示	√	√
历史积水数据正常显示	√	√

7.4 温度监控功能测试

本部分主要测试的内容是温度监控功能的测试标准和测试结果。

表 7-4 报警配置功能测试表

要测试的功能	预期结果	实际结果
折线图实时显示温度数值	√	√

7.5 湿度监控功能测试

本部分主要测试的内容是湿度监控功能的测试标准和测试结果。

表 7-5 湿度监控功能测试表

要测试的功能	预期结果	实际结果
折线图实时显示湿度数值	√	√

7.6 积水监控功能测试

本部分主要测试的内容是积水监控功能的测试标准和测试结果。

表 7-6 积水监控功能测试表

要测试的功能	预期结果	实际结果
折线图实时显示积水数值	√	√

7.7 车位和数据功能测试

本部分主要测试的内容是车位和数据功能的测试标准和测试结果。

表 7-7 车位和数据功能测试表

要测试的功能	预期结果	实际结果
3 个车位没车时正常显示	√	√
3 个车位有车时正常显示	√	√
总览环境信息	√	√

7.8 本章小结

本章主要是对系统的功能进行测试，包括用户管理测试、报警配置测试等，系统内每一项测试切入点都能够正常按预期结果实现通过本次测试，我们验证了基于物联网的停车场监控数据采集系统的各项功能均符合设计要求，能够在实际应用中发挥重要作用。

结 论

本论文旨在研究并实现基于物联网的停车场监控数据采集系统。该系统以 STM32 单片机为核心控制器，结合空气温湿度传感器、液体传感器以及红外传感器，实现了对停车场环境信息和车位状态的实时监控。同时，通过 OLED 屏幕实现了数据的本地显示，并利用 MQTT 协议和通过通信猫云平台实现了数据的远程传输和转发。

在系统的设计和实现过程中，我们首先完成了硬件平台的搭建，包括 STM32 单片机的选型、传感器的连接与配置，以及 OLED 屏幕的驱动与显示。接着，我们设计了系统软件，包括数据采集程序、数据处理程序、数据显示程序以及 MQTT 通信程序。通过不断的测试和优化，我们确保了系统的稳定性和可靠性。

通过实际测试，我们验证了系统的各项功能。系统能够准确检测停车场的空气温湿度、积水情况以及车位状态，并将这些数据实时显示在 OLED 屏幕上。同时，系统能够通过 MQTT 协议将数据上传到通信猫云平台实现数据的远程转发。这为停车场管理提供了有效的数据支持，有助于提升停车场的管理水平和用户体验。

此外，我们也对系统的性能进行了测试。结果表明，系统在响应速度、稳定性、准确性以及数据传输效率等方面均表现出良好的性能。系统能够在短时间内快速响应环境变化，并实时更新显示数据。同时，系统能够长时间稳定运行，无异常情况发生。

综上所述，基于物联网的停车场监控数据采集系统实现了对停车场环境信息和车位状态的实时监控和远程传输，具有较高的实用性和可靠性。该系统不仅为停车场管理提供了便捷的数据支持，也为物联网技术在停车场管理领域的应用提供了有益的探索和实践。未来，我们可以进一步优化系统性能，扩展系统功能，以适应更多复杂和多样化的应用场景。

致 谢

时光匆匆,我的大学生活即将迎来终点。回首过去的日子,我深感自己收获颇丰,同时也感慨良多。在此,我要向所有给予我关心、支持和帮助的人表达我由衷的感激。

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附录 A 译文

基于雾计算和物联网的智能停车系统的设计与开发

摘 要

目前，停车系统采用单一网关为中心的解决方案，即云的数据处理，导致单点故障，数据丢失和高延迟的可能性。此外，停车位选择过程考虑的标准，最大限度地提高停车利用率和收入。由于采用了固定定价策略，定价策略不能实现高收入。为了解决这些问题，本文提出了一个基于物联网(IoT)的智能停车系统，该系统为司机和停车场管理员提供有用的信息，包括可用停车位和相关服务，如停车导航、预订和可用性评估。提出了一种由多个传感器节点和雾云计算层组成的多层体系结构。获得的停车数据通过雾计算节点进行处理，以便获得所需的实时停车数据。提出了一种基于最短到达时间的最优停车位选择算法。进行了概念验证和仿真评估，以验证系统的性能。研究表明，与现有的停车系统相比，该系统可以减少 16-46% 的停车到达时间。此外，停车管理局的收入增加了 10-15% 。

关键词 智能停车；定价策略；停车位选择；可用性估计；雾计算

第1章 介绍

智慧城市概念整合资讯及通讯科技，以促进可持续发展、经济增长及改善生活质量。智慧城市的其中一个发展方向，是发展智慧交通及有效率的交通管理系统，以便优化驾驶人士的出行计划及纾缓交通堵塞。目前，司机们通过搜索所有的停车位，直到找到一个可用的停车位，从而获得一个自己的停车位。这会导致浪费宝贵的时间，从而阻碍经济增长。此外，由于交通堵塞等待时间过长，车辆不必要地燃烧燃料，导致温室气体排放量。这些问题可以通过使用智能停车系统来解决，该系统可以帮助司机在到达目的地之前确定和预定停车位。这种系统在上世界上几乎每一个主要城市都会变得非常重要，特别是在大型活动期间（例如，会议、宗教活动、体育活动、节日和音乐会）。它们将有助于解决交通堵塞问题，减少污染，并最大限度地减少对城市景观和环境的负面影响。

智能停车系统是交通管理策略的一部分，通过帮助车主有效地停车，减少停车到达时间和交通堵塞，来管理停车流程。智能停车系统的发展由于其对经济、环境和美学的影响，已经引起了学术界和工业界研究人员的关注。物联网（IoT）正在成为智能对象通信的领先技术。物联网是智能停车系统的关键技术，它采用了不同的传感技术（如超声波、磁强计和视觉传感器）和网络技术（如 ZigBee、Wi-Fi 和蜂窝网络）。它通过收集实时传感器数据并与机器学习、传感器融合，和计算机等其他技术相结合，促进了实时监测和控制。一些商业停车系统已经被开发出来，如智能停车，PlacePod 和 SitrafficScala。这些系统提供了一些功能，比如远程停车预订、费用支付、交互式停车地图等等。然而，它们的主要缺点是公共开发人员的开发成本高，开发能力有限，因为它们不是开源的。此外，目前的系统不能满足今天的停车需求和智能城市发展的期望。

当前的系统使用一个以网关为中心的云计算解决方案。物联网数据被收集并发送到互联网进行处理，这导致了单点故障的可能性，网络中断时数据丢失，以及由于数据通过网络流向云而导致的高延迟。此外，大多数实施的停车监测系统是本地操作的，并不向司机提供在哪里可以找到停车位的实时信息。所使用的通信技术是基于有线系统或短距离通信方法，如 Wi-Fi、蓝牙或 ZigBee，这在室内和室外停车场的大多数情况下是不可靠的。它们还面临着许多与干扰、能源消耗和资源有限相关的问题。

此外，停车位的选择也考虑了不同的标准，如用户和运营商的偏好。现有的系统只考虑需求和驱动程序首选项。选择决策必须基于一个标准，减少拥堵，最大限度地利用停车场，因此，收入。另一方面，停车场的定价策略通常是基于短期停车的固定每小时费率和长期停车的固定订购费率，这并不能获得很高的收入。根据先前的入住率周期性地得到固定的入住率。这种策略没有利用停车的可用性和需求，因此它导致次优定价。因此，本文报道了一个基于多层物联网体系结构的智能停车系统的设计与开发，以解决上述问题。该系统考虑不同的停车计划，包括有或没有提前预定的短期停车计划和长期停车计划。该系统是通过实施和模拟的选定区域代表我们的案例研究（即 Kudai 停车场）进行评估。该系统具有以下特点：

（1）一个由物联网、雾和云层组成的多层物联网架构，以克服单一网关为中心的解决方案的局限性。分布式雾计算节点用于本地处理数据，并将基本数据转发给云服务器。传感器节点通过节能的 LoRaWAN 通信技术连接到雾节点，视觉传感器通过 Wi-Fi 技术连接的除外。这是因为 LoRa 通信提供了较低的数据速率，适用于不连续发送大量数据的传感器节点。

（2）一种停车位选择算法，该算法将停车位到达时间作为停车位选择准则，以减少等待时间和拥挤，从而减少空气污染。考虑了停车场的到达信息，包括进出停车场的速度、停车可用率和车辆的 GPS 定位。同时利用路口处的视觉传感器信息获得拥塞率。

（3）以最大限度地利用停车场为目标的便于停车预定过程的停车场可用性估计。提前至少三个小时预约可以让不预约的顾客有机会找到停车位，从而减少停车场入口处的交通堵塞。

（4）使停车收入最大化的混合定价算法。该算法利用了短期计划的小时费率定价策略和长期计划的动态定价策略。动态定价表示与需求和收入相对应的动态订阅率。

论文的其余部分组织如下。第二部分讨论了智能停车系统的相关研究，并强调了它们的局限性和我们提出的系统的贡献。第三部分介绍了所提出的停车系统，包括所提出的多层物联网体系结构和操作流程、停车位选择、停车可用性估计和混合定价算法。第四部分详细介绍了所考虑的案例研究的实现和仿真评价，以及所使用的硬件和

软件工具。并对所得结果进行了讨论和分析。最后，在第 5 节中提出了结论和对未来工作的建议。

第2章 相关工作

文献中已经提出了几篇综述文章，探索、分类和讨论智能停车解决方案的技术方面。这一部分讨论了现有研究中的相关智能停车系统，这些研究是通过在 ISI Web of Science 和 Scopus 等数据库中使用关键词“智能停车”和“物联网停车”进行搜索获得的。使用这些关键词搜索可以得到 316 篇文章，其中 127 篇来自 ISI Web of Science，189 篇来自 Scopus。从这两个数据库中删除了重复的文章，产生了 275 篇最终文章。图 1 显示了用于获取所考虑的相关文章的评审方法。

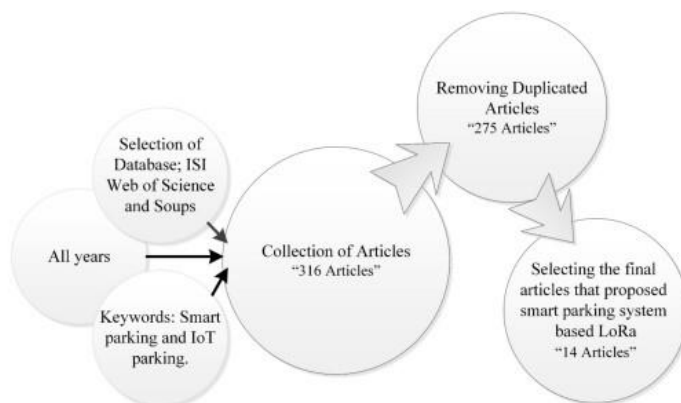


图 1 相关文章选择方法

这些文章的范围涉及不同的问题，包括应用程序和可用性、定价和合同、可用性预测和分配等。根据问题范围、解决方法和评估方法，对 275 篇文章进行了进一步的研究。应用审查方法有助于摘录与本文的范围和目标最相关的 14 篇文章。不同的标准可以考虑分类系统，包括停车环境，提供服务，或核心传感技术。停车场系统可以分为基于核心电子传感技术的无线传感器系统，基于运动传感器的系统，和基于摄像头传感器的系统。本节将仅讨论与本文中提出的解决方案最相关的智能停车解决方案，即基于无线传感器的系统，利用 LoRa 作为系统的网络基础设施。

一个具有 LoRa 基础设施的智能停车系统的概念验证实现已经在几篇论文中提出。实践表明，该系统在能量、完整性和可扩展性方面对大城市的停车监控和管理具有较高的效率。在中，该系统是通过在线预订系统在哈桑努丁大学的停车场实施的。类似地，作者开发了一个基于 LoRa 的智能停车系统，该系统包含一个传感器模块、一个云服务器模块、预订应用程序和一个第三方支付平台。该系统在中国浙江省的两个城市进行了测试：宁波和舟山。实验结果显示，在监控和预订能力的增强。然而，

LoRa 通信的质量,安全性和可靠性没有在复杂的城市环境中的真实停车场进行测试。

智能停车解决方案提出,它结合了 libel 智能停车传感器,lorawAN,以及 Kubernetes 集群与 MQTT 和 MongoDB。通过克服信息共享瓶颈,这种组合增强了智能停车系统的可用性、可扩展性和可移植性。如果用于信息收集的主服务器失败,系统将在消息传递服务器中累积所有消息,以便进一步处理。另一个智能停车解决方案与定价算法的收入最大化被提出在。价格是动态设置的,以平衡可用和请求的停车位,并确保收取最低停车费。该解决方案包括多个传感器,识别车辆,并获取上下文和环境信息。此外,利用边缘云计算最小化网络负载和网关数量。

其他研究已经提出了提高停车解决方案互操作性的方法。系统采用了以物联网网关为中心的架构,允许所有支持的传感器设备和/或执行器之间通过使用 LoRa 网络的 LoRagateway 进行实时交互。网关依次将数据包转换为服务器可接受的格式。因此,不需要网络服务器将数据包路由到它们的目的地。然后,接收到的数据包被 Spring 函数解析,并在需要时用数据库中的数据进行处理,以最终确定结果并将其传递给用户。停车系统的支付功能利用 API 支付高速公路。一个新的基于框架的 Wise-IoT 系统被开发用于物联网应用程序和设备的全球互操作性和移动性。该系统基于不同的 IoT 基础设施(一个是 LoRaWAN)和两个不同的互操作层进行开发。该系统在世界两个城市实施:釜山和桑坦德。在桑坦德生成的传感器观测结果发送到一个 M2M 和 FIWAREIoT 平台的不同实例上,而在釜山,它们分布使用一个 M2M 作为后端。因此,为桑坦德银行开发的智能停车场可以在釜山的不同地点使用,反之亦然。

为了在停车系统中保护隐私,作者提出了一个增强的 LoRaWAN 安全协议,该协议提供了基本的连接功能,并避免了安全问题。它防止恶意网络服务器破坏设备与其应用程序服务器之间的端到端安全性。使用不同的工具对该协议进行了分析,结果表明该协议克服了网络延迟和信令开销方面的两个握手选项。提出了一种使用安全元素来安全存储密钥和处理 LoRaWAN 协议的新方法。此外,基于一个安全元件,提出了一种高效的可执行文件完整性验证方法。表 1 总结了最相关的研究。

表 1

Research Study	Online Platform	Parking Navigation	Reservation Service	Parking Availability Estimation	Payment Platform	Tested Parking Area
Angelov, K.K. 2020.	✓	✓	x	x	x	None
Anshar, M., et al., 2020.	✓	✓	✓	x	x	Campus parking
Shi, J., et al., 2017.	✓	✓	✓	x	✓	City parking
Barriga, J.J., et al., 2020.	✓	✓	✓	x	✓	City parking
Sotres, P., et al., 2018.	✓	✓	x	x	x	City parking
Proposed System	✓	✓	✓	✓	✓	City parking

第 3 章 建议的智能泊车系统

提出的智能停车系统监测和管理停车场，使车辆停放过程更加方便和舒适，引导司机到他们的停车位，减少拥堵和更短的行驶时间。该系统由云计算、雾计算节点、一组传感器和停车管理软件组成，用于获取和处理与停车预订、可用性和定价相关的数据。这些组件具有将数据发送到雾节点的无线连接能力，雾节点处理收集到的数据。该系统根据停车计划考虑三个停车等级：(i)有停车预约的车辆，(ii)有预约停车的车辆，以及(iii)没有预约停车的车辆。下面的小节详细介绍了停车系统的体系结构和操作、停车可用性估计、停车位选择和定价算法。表 2 列出了整篇论文中使用的表示法。

表 2 注释清单

Symbol	Definition	Symbol	Definition
i	Edge lane	λ_i	Traffic density on that edge
V	Vehicle number in an edge	M_i	Maximum capacity of vehicles on that edge
C	The total congestion rate of all edge lanes on the route	T	The total number of all edge lanes on the route
S_i	A parking spot	S_i	The current state of parking spot S_i
P_i	A parking lot	Cap_i	Capacity of a parking lot
d	Discount percentage	r	Range of hours
P_T	The total parking price	P_T	The first-hour price rate
P_s	The price rate after the first hour	h	Total parking hours
N	The maximum value of ranges	R	The parking revenue
k	The number of available parking spots for subscriptions	P_s	The subscription price rate of n th parking lot.
U_n	The number of clients who afford a specific rate	$Rate_n$	The maximum rate that n th driver affords

3.1 停车系统体系结构与操作 workflow

停车系统的多层物联网架构有效地管理和监控停车场，并保持较低的拥堵率。该体系结构克服了传统传感器云体系结构的局限性，利用分布式雾计算节点对数据进行局部处理，只将关键数据转发给云服务器，而不是将所有数据转发给云服务器。这仅仅是为了保持所需数据的快速处理和传输，从而实现实时应用体验，将一般视图或长期服务(例如，不同位置的停车场或历史停车数据)所需的数据传输到云端。该架构由三层组成：(i)物联网停车层、(ii)雾层和(iii)云层，如图 2 所示。

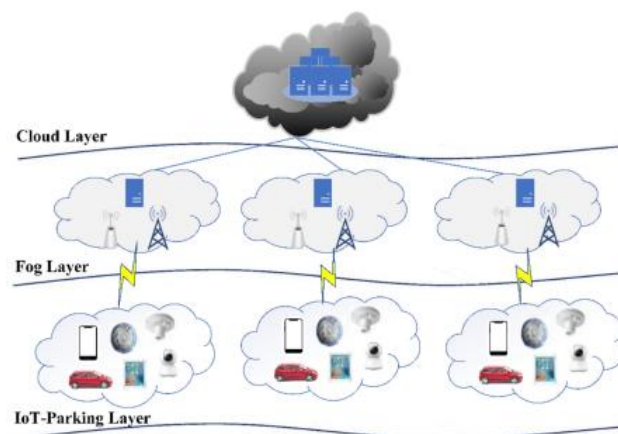


图 2 多层停车系统体系结构

3.1.1 物联网停车场

最底层由一组传感器节点组成，通过检测车辆和收集有关停车区域的数据来监控停车场。不同类型的传感器可以被利用，放置在地面（例如，电磁传感器），在天花板，或在街柱（例如，超声波和视觉传感器）。根据停车场及其周围环境，得到最适合利用的传感器。磁强计传感器和热成像仪被用于暴露在不同天气条件下的停车场，以探测磁场的变化和温度差异。视觉传感器是首选的更复杂的停车区域或如果他们已经安装。

在这一层，我们的系统采用了多种传感器和数据采集设备，包括超声波、磁场和视觉传感器、射频识别 (RFID)、全球定位系统 (GPS) 设备、二维码阅读器和停车场管理平台。例如，视觉传感器被放置在入口、出口和交叉路口，为停车位的选择提供有用的信息，并监测违反停车规则的情况。与蓄电池相连的太阳能电池板用于提供数据采集设备所需的电力。传感器节点通过节能的 LoRaWAN 通信技术与雾节点进行通信，而视觉传感器通过 Wi-Fi 技术进行通信。这是因为 LoRa 通信提供了较低的数据传输速率（即高达 37.5 kbps），适用于不连续发送大量数据的传感器节点。LoRa 技术以 25mW 的最大传输功率促进长达 5 公里的远程通信。根据当地的规定设置了 LoRa 的结构参数和频率载波。

3.1.2 雾层

正如前面提到的，这个层的目的是提供一个强大的解决方案，通过将一些处理/分析（例如，数据预处理、过滤和压缩）迁移到靠近低层次层的地方来实现实时控制和

分析需求。这包括停车位的选择，停车可用性估计的过程，以及我们提出的停车系统的定价。雾层通常由几个雾节点组成，其中根据其需求，特定服务的进程与特定节点相关联。雾节点定义为具有处理能力的分布式计算实体，其中每个节点可由一个或多个物理设备（例如网络设备、专有服务器或计算服务器）组成。如果存在高执行负载或连通性问题，每个节点可以同时处理多个传感器节点，并且可以进行传感器节点切换。此外，协同任务分配可以在雾节点之间共享计算负载，以减少总体运行延迟。

3.1.3 云层

最高层实际上提供了无限的存储和计算资源。云层为提议的停车系统提供了多种服务，例如使用分析、统计和历史信息、收入监控以及自动支付和管理。云层还支持安全、实时的在线停车预订信息，以及通过网上运行或安装在移动设备上的停车平台指向预订停车位的方向。云层可以采用加密技术隐藏支付计划和驱动程序信息，以保证安全和隐私。此外，历史和长期信息可用于大数据分析，以加强交通管理和智能城市解决方案，以及实时反馈通知和紧急警报。

拟议停车系统的工作流程如图 3 所示。该过程首先通过传感器节点和停车管理平台获取停车场数据。数据采集后，车辆的物理变量被发送到雾节点，包括停车预定、到达和车辆检测。雾节点每 10 毫秒处理一次收集到的数据，以确保准确的决策，最大限度地减少资源的使用和处理能力。处理获得的数据以确定关于到达、停车可用性、停车匹配和停车预定的信息。在此基础上，实现了停车可用性估计、安全警报、动态定价和停车位选择四种算法。下面的小节将对这些算法进行详细的描述。算法的结果通过网络设备传输到云计算层，从而将停车信息传输到停车管理平台。

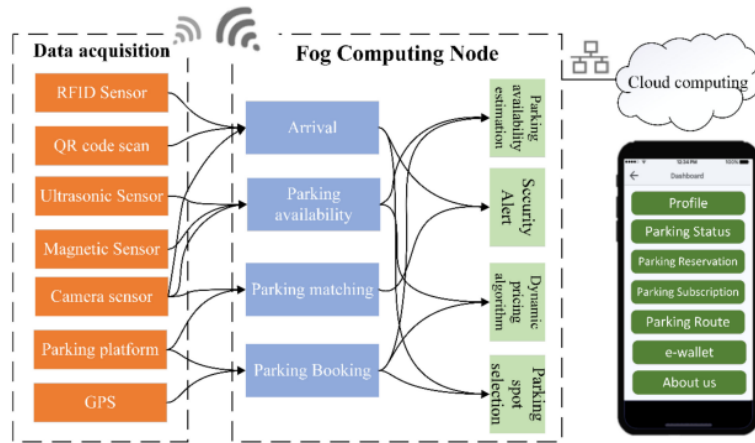


图 3 智能停车系统的运行

3.2 停车位选择算法

停车位的选择是在车辆进入停车场时开始的。停车位选择过程使用优化算法，以最小化到达停车位的时间为目标，选择最优停车位。该算法适用于有或无停车预约的车辆，但不适用于有停车预约的车辆。它利用停车到达和预订的信息，包括进出速度，停车可用率和车辆的 GPS 位置。此外，路口的视觉传感器信息将使用提供的拥挤模型来获取拥挤率。入口点作为一个导航启动点来选择停车场，停车场是在车辆到达时确定的。该算法执行一个前向搜索过程，其中入口点的位置，然后计算到达一个可用停车位的预期时间。此外，还进行了回溯处理，计算从停车位回到前一入口点的到达时间。

停车位选择过程的一个示例如图 4 所示。每条道路有两条边缘车道和最大车辆通行能力。对于边缘车道 i ，交通流密度 λ_i 计算为

$$\lambda_i = V/M_i \quad (1)$$

其中 V 代表边缘车辆的数量， M_i 代表边缘车辆的最大容量。密度以百分比的形式显示在图 4 中。路线上所有边缘车道的总拥挤率计算为

$$C = \sum_{i=1}^j \lambda_i/T \quad (2)$$

其中 T 是路线上所有边缘车道的总数。寻找最佳停车位的过程从给定的入口节点开始，到达达到最短行车时间的可用停车位。因此，搜索图由一个作为入口点的根节点和下一级节点组成，下一级节点是所有可以直接从上一级（即第一个根节点）访问的节点，以此类推，直到确定候选可用停车位。该图形构成一个树状视图，其中最大

深度表示该深度内的路口数和候选停车路口数。使用 Dijkstra 算法可以找到具有最短旅行时间的最优路径。

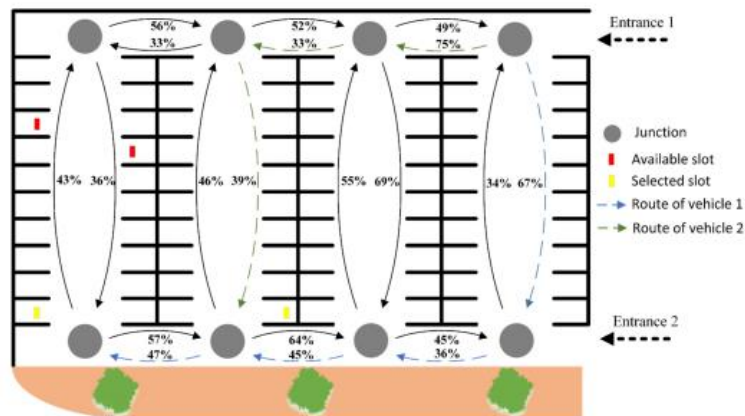


图 4 停车位选择与车辆路线

3.3 停车场可用性评估

为了方便停车预定过程，提出了停车可用性估计方法。如果预计停车位将在要求的时间可用，则允许提前至少三个小时预订。这样可以最大限度地利用停车场，让没有事先预定停车位的车辆有机会找到停车位，从而减少停车场入口处的交通堵塞。系统中的雾节点知道当前时刻的停车可用性。然而，这一信息不足以估计近期内的泊车可用性，因为车辆到达率在不同的季节期间波动。利用历史季节性数据的预测分析可以预测交通的波动，这些数据是从停车场入口和交叉路口的传感器收集的。这些历史数据存储在云中，并根据季节划分为若干实例。停车周期必须足够小（例如，一分钟），以尽量减少停车动态的变化。这些数据被用来建立一个基于马尔可夫过程的随机模型，表示随机变化的可用停车位。

马尔可夫模型从相似时期的历史停车可用性信息推断未来时期的停车可用性。该模型获得了每个停车场的几个参数，包括观察时间窗口、到达率、出发率和状态转移矩阵。表 3 显示了停车场数据的一个例子，其中对类似季节的多历史一分钟观测值进行了平均。这个周期设定得很短，以便在固定到达率和离开率的五分钟时间窗口上获得停车场的离散状态变化。选择短期的原因是为了证明每个观察窗口内的到达和离开率是固定的这一假设是合理的；然而，到达和离开率的变化是跨观察窗口考虑的。每个观察窗口的大小可以得到一个代表停车动力学信息的单一状态转移矩阵。

表 3 停车场数据样本

Parking Lot p_i				
¹ Season				
Year	Month	Day	Period (Time)	Available Spots (Average)
1439	Safar	5	9 h 56 m	987
1440	Safar	5	9 h 56 m	970
1441	Safar	5	9 h 56 m	950
1442	Safar	5	9 h 56 m	987
1443	Safar	5	9 h 56 m	960

对于每个停车场，停车状态被确定，其中每个状态代表停车可用范围。利用经典的马尔可夫链模型和肯德尔公式得到了停车位的未来可用状态 P ，其中 S_{jis} 的当前状态。该泊松分佈在模拟中被用来表示在一个单一观察期内进出停车场的车辆数量的固定平均值的周期内的到达过程。建议的停车可用性估计使用一个停车场的五个状态范围，从高度占用状态 S_1 到高度可用状态 S_5 。图 5 显示了容量为 $C_{api} = 1000$ 的停车场的状态转换矩阵的示例，该矩阵来源于历史数据，观察时间为 5 分钟。一个归一化转移矩阵的概率权重也如图所示。

算法 1: 混合定价算法

```

BEGIN;
If short-term parking plan is selected then
    Š Process pricing for short-term parking plan;
else if long-term parking plan is selected then
    Š Process pricing for short-term parking plan;
End;
    
```

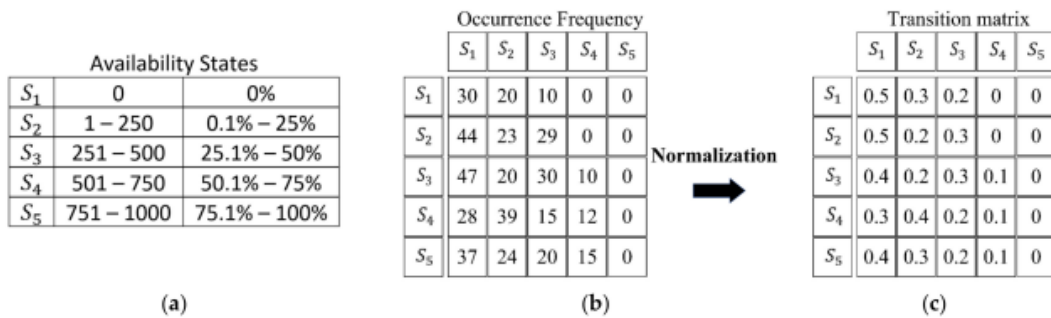


图 5 马尔可夫模型：(a) 可用性状态、(b) 出现频率和 (c) 转移矩阵

3.4 混合定价算法

开发了一个算法来促进两种停车计划的定价策略：(i) 短期停车计划 (即有预订和无预订的停车) 和 (ii) 长期停车计划 (即每月和每年的停车订阅)。混合定价包括短

期计划的小时费率定价策略和长期计划的动态定价策略。算法 1 给出了混合定价算法的伪码。

每小时收费率考虑指定某些折扣率的可能性，它为第一个小时设定一个固定的价格率，然后根据在超过指定的小时 (r) 范围时采用指定折扣百分率 (d) 的范围计算较后小时的价格率。停车费总额计算如下：

$$PT = PF + Ps(h - 1) - XdPs(h - 1) \quad (3)$$

其中 PT 是总停车价格，PF 是第一个小时的价格，Psis 是第一个小时后的价格，h 是总停车时间，X 的计算方法如下：

$$X = \text{Min}(bh/rc, N) \quad (4)$$

其中 N 是范围的最大值。例如，如果总共需要停车 20 小时，第一个小时的停车费是 10 美元，第一个小时之后的停车费是 5 美元，如果每个车程包括 3 小时，每个车程有 10% 的折扣，总的停车费将是

$$PT = 10 + (5 * 19) - (5 * 0.1 * 5 * 19) = \$52.5$$

对于长期停车(即，订阅)，一个动态订阅率是分配对应的需求和收入。该策略的目标是优化停车收入 $R = \sum_{k=1}^n P_n$ ，其中 k 是可供订购的停车位数，Pn 是第 n 个停车场的订购价格。这可以通过在选定的季节期间进行一次或多次招标来实现。首先将最大投标价值分配为价格比率，然后将其降低到与需求和收入相匹配的比率，从而得到认购价格比率。例如，在 20 个客户每月出价 200 美元，3 个客户每月出价 150 美元，7 个司机每月出价 90 美元的情况下，最佳定价率将是每月 150 美元。为确保公平，停车场管理部门将预先规定最高投标价格。在订阅价格率设定后，能够支付价格的客户将根据他们的出价价值和时间戳按递减顺序排列。优化算法和最大化目标可以表示为

$$\text{Maximize } R \text{ Subject to : } k \leq \text{Un Ratem} \geq P_n ; \forall n \leq k \quad (5)$$

其中 Ratem 是 mth 驱动程序将支付的最大利率。排序算法合并排序或堆排序可以用来搜索最佳的最大值，因为它们的复杂性较低。

第 4 章 执行及评估

为了评估拟议的智能泊车系统，我们考虑了邻近 theHoly Mosque 的麦加停车场作为个案研究。具体来说，是选择 Kudai 的停车场进行模拟和实施。麦加有 13 个停车场，总面积达 150 万平方米，最多可容纳 4.5 万辆车。古代停车场面积约 74,000 平方米，停车位 2500 个。图 6 显示了停车场和运输站，包括古代停车场。在朝觐和朝觐期间，交通高度拥堵，但预计该系统将减少交通堵塞，尤其是在停车场入口处。根据一般统计机构的数据，2019 年麦加朝圣季节进入麦加的车辆数量约为 32978 辆。沙特的愿景是，到 2022 年，朝觐和朝觐季节的朝圣者总数将增加一倍，这反过来需要更好的交通和停车管理。

为了验证这个概念，我们选择了 Kudai 停车场部分地方的一个简化版本进行试验性评估，如图 7 所示。请注意，一些组件没有显示在图中，因为他们放在后面的停车场。为了简单起见，概念验证实现中没有包含可视化传感器。实验的硬件组件包括 HC-SR04 超声波和地磁传感器，一个 RFID-RC522 阅读器模块，一个 S50 RFID 卡，一个 Adafruit Feather 32u4 with LoRa 无线电模块，一个可充电的锂离子电池，一个小型光伏面板，LED，树莓 Pi 3 和一个 HP 工作站。

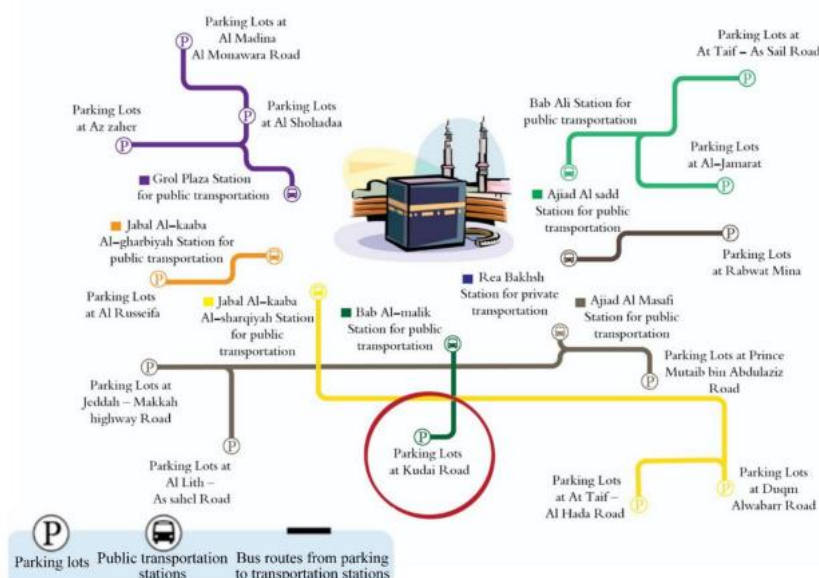


图 6 麦加神圣清真寺附近的停车场和交通站

一块带有 868/915 MHz 无线电模块的羽毛 32u4 LoRa 板被用于 LoRa 通信。超

声波、地磁和射频识别传感器的系列 RX/TX 引脚连接到 Adafruit 翎上的 Arduino 系列 RX/TX 引脚。数据从 Arduino 系列 RX 中读取并通过无线电传输。数据由机载 LoRa 无线电接收，并通过另一个串行端口发送到覆盆子圆周率，它在实验中被用作雾节点。节点在本地收集、处理和排序数据，并将数据转发到后端服务器。它处理停车位选择、停车可用性估计和混合定价算法。运行 Ubuntu 的 HP 工作站被用于后端分析，以监控和分析网络的整体性能。它处理历史数据来估计停车可用性，并相应地允许或不允许停车预定。通信模块和传感器通过一个可充电的锂离子电池提供电力，该电池由一个紧凑的光伏面板充电。Blynk 应用程序可以用来显示在雾节点中执行的算法的输出。

通过仿真评估，研究了该停车系统的优点，特别是车位选择和混合定价算法。自由开源模拟器，城市机动性模拟 (SUMO) 被用来明确地定义每一辆车与地图上的特定路线。SUMO 中车辆的默认行为是以最高指定速度行驶并避障。它还包含一个停车包来定义一个停车区域，并将车辆引导到一个特定的停车点。模拟的 Kudai 停车场区域取自 OpenStreetMap (OSM)，并插入到 SUMO 中以表示模拟区域。图 8 显示了 SUMO 模拟器中的 Kudaipark 停车场。



图 7 Kudai 停车场和概念验证设置

我们将该系统与传统的泊车方法作比较，传统的泊车方法是由现时的位置开始寻找可供使用的泊车位，直至找到为止。我们考虑了一个最多可容纳 150 个泊车位的停

车场，其中每分钟有 10 部车辆抵达，每分钟有 5 部车辆离开。最初的停车可用性是随机产生的均匀分布与 90% 的最大率。如果超过 80% 的停车位被占用，停车场就被认为是满的。用于模拟评估的测量性能指标是到达时间、搜索时间和停车收入。到达时间是车辆从入口到达指定停车位所花费的时间，而搜索时间是搜索停车位所花费的时间。停车收入，另一方面，是总收入来自所有停车场与不同的停车计划在一个停车场。

图 9 显示了使用建议的系统 and 传统方法考虑不同拥堵率和可用停车位的到达时间。模拟被执行直到一个可用的点被获得。图 9a 显示了只有一个可用停车位的到达时间，拥堵率介于 0.1 和 0.9 之间。结果表明，使用该算法的到达时间明显少于传统的停车位选择方法。传统的搜索方法比提出的算法多花费 20% 到 60% 的时间。虽然提出的算法总是优于传统的方法，两种方法之间的到达时间差减少，拥塞率较高，因为车辆很难在所有的路口快速移动。在图 9 中，到达时间显示了 1 到 10 个可用停车位和 0.8 的固定拥挤率。在所有情况下，该算法的到达时间均低于传统算法。当只有一个停车位可用时，该算法的到达时间明显低于传统方法，即盲目搜索区域内所有的停车位，直到找到一个可用的停车位。然而，随着停车位数量的增加，到达时间之间的差异逐渐减小，因为可以找到更多的可用停车位。

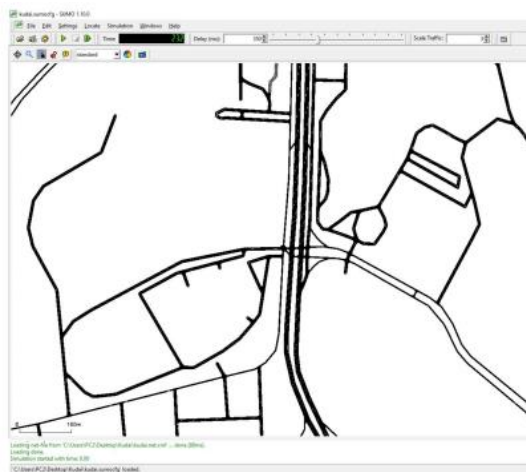


图 8 SUMO 模拟器中的 Kudai 停车场

图 10 显示了搜索时间，其中包含 1 到 10 个可用停车位，固定拥挤率为 0.8。结果表明，由于搜索时间是到达时间的大部分，因此搜索时间度量和到达时间具有相似的模式。该算法具有较低的搜索时间和到达时间，当只有一个停车位可用时，搜索时

间和到达时间显著降低。当有更多的停车位时，这两种方法之间的差异变得更小。

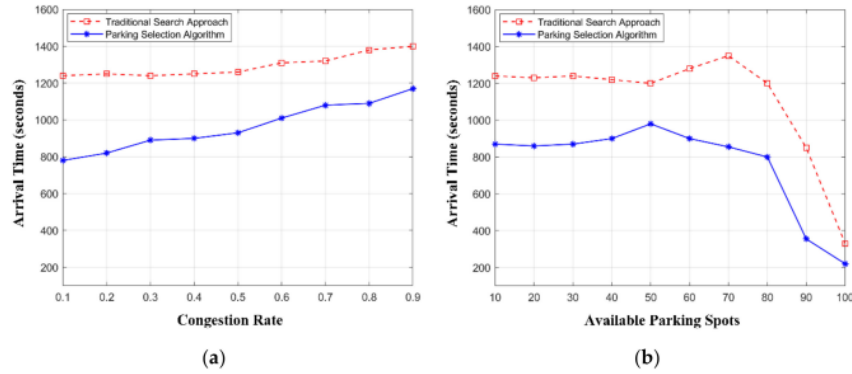


图 9 利用停车选择算法和传统的搜索方法，考虑到：(a) 不同的可用停车位，(b) 不同的拥挤率

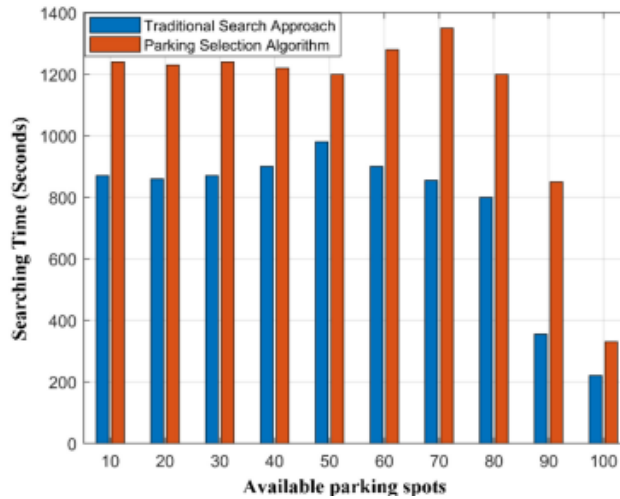


图 10 基于停车位选择算法和传统搜索方法的不同可用停车位搜索时间

考虑不同的停车预订请求，评估混合定价算法和固定定价算法的收益。在这里，我们考虑了一个最多可容纳 150 个停车位的停车场，其中 50 个停车位用于长期规划，100 个停车位用于短期规划。此外，可供认购的停车位数量被认为少于可以支付违约认购费用的客户数量。短期计划的固定定价为每小时 10 美元，长期计划的固定定价为每月 2000 美元。对于短期计划，动态定价为第一个小时 10 美元，其余小时有 10% 的折扣，而对于长期计划的投标率随机高于违约率。图 11 显示了与固定价格相比，所有停车计划(包括短期和长期停车计划)的总收入。两种停车方案的占用率从 10% 到 100% 不等。动态定价方案使长期计划收入显著增加，但由于折扣，短期计划收入小于固定定价收入。

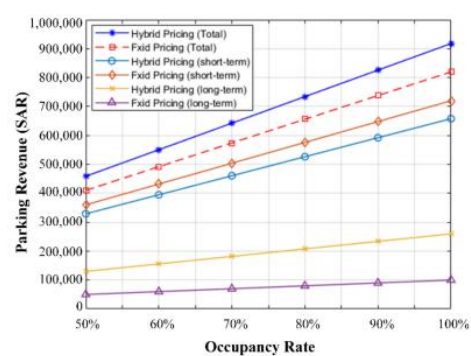


图 11 停车场一个月的收入与占用率不同

结论

智慧城市旨在改善居民在交通和旅游服务等不同生活方面的生活质量。在这种情况下，智能停车系统帮助司机提前获得和预订停车位。为了克服现有停车系统的局限性，提出了一种基于 LoRa 通信接口的多层物联网结构的智能停车系统。该系统考虑了不同的停车计划，包括有预订和无预订的短期停车计划和长期停车计划。采集到的停车数据通过雾计算节点进行处理，以便于获得所需的实时停车数据。此外，还引入了一种动态定价算法，目的是使停车场管理机构的收益最大化。提出了一种最小到达时间最优停车位的求解算法。建议的系统是通过实施和模拟的一个选定的区域代表我们的案例研究(即，库代停车场)进行评估。结果表明，停车系统提高了停车到达时间和收益。在未来的工作中，我们计划开发和安装一个测试领域的建议，在我们的案例研究停车系统。

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利益冲突：作者声明没有利益冲突。

附录 B 外文原文

Article

Design and Development of Smart Parking System Based on Fog Computing and Internet of Things

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Abstract: Current parking systems employ a single gateway-centered solution (i.e., cloud) for data processing which leads to the possibility of a single point of failure, data loss, and high delays. Moreover, the parking-spot selection process considers criteria that do not maximize parking utilization and revenue. The pricing strategy does not achieve high revenue because a fixed pricing rate is utilized. To address these issues, this paper proposes a smart parking system based on the Internet of Things (IoT) that provides useful information to drivers and parking administrators about available parking spots and related services such as parking navigation, reservation, and availability estimation. A multi-layer architecture is developed that consists of multiple sensor nodes, and fog and cloud computing layers. The acquired parking data are processed through fog computing nodes to facilitate obtaining the required real-time parking data. A novel algorithm to obtain the optimal parking spot with the minimum arrival time is also presented. Proof-of-concept implementation and simulation evaluations are conducted to validate the system performance. The findings show that the system reduces the parking arrival time by 16–46% compared to current parking systems. In addition, the revenue is increased for the parking authority by 10–15%.

Keywords: smart parking; pricing strategy; parking-spot selection; availability estimation; fog computing



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1. Introduction

The smart city concept integrates information and communication technologies (ICT) to facilitate sustainable development, economic growth, and quality-of-life improvement [1–3]. One of the directions leading towards smart cities is the development of intelligent transportation and efficient traffic management systems to optimize drivers' travel planning and alleviate traffic congestion [4–7]. Currently, drivers obtain a parking spot themselves by searching all spots until an available spot is found. This leads to a waste of valuable time that may hinder economic growth [3]. In addition, during the waiting time due to congestion, the vehicles are burning fuel unnecessarily, causing high greenhouse gas emissions [8]. These issues can be addressed using a smart parking system that helps drivers to determine and reserve parking spots in advance before reaching their destination. Such systems would become highly significant in almost every major city in the world, especially during large events (e.g., conferences, religious events, sporting events, festivals, and concerts). They would help in tackling traffic congestion, reducing pollution, and minimizing the negative impact on the city landscape and environment [9].

Smart parking systems form part of traffic management strategies and manage parking processes by helping vehicles to park efficiently in a way that reduces parking arrival time and traffic congestion. The development of smart parking systems has attracted the attention of researchers in academia and industry, due to their economic, environmental, and aesthetic impacts. The Internet of Things (IoT) is emerging as a leading technology for smart object communication [10]. IoT is the key enabler technology for smart parking systems, which employ different sensing technologies (e.g., ultrasonic, magnetometers, and visual sensors) and networking technologies (e.g., ZigBee, Wi-Fi, and cellular). It facilitates real-time monitoring and control by collecting real-time sensor data and integrating with other technologies such as machine learning, sensor fusion, and computing [11]. Several commercial parking systems have been developed such as SmartParking [12], PlacePod [13], and Sitraffic Scala [14]. These systems offer several features such as remote parking-spot booking, fee payment, interactive parking maps, and others. However, their main drawback is the high cost and the limited development capability for public developers since they are not open source. In addition, the current systems do not meet today's parking needs and the expectations of smart city evolution.

The current systems employ a single gateway-centered solution using cloud computing. IoT data are collected and sent to the Internet for processing, which leads to the possibility of a single point of failure, data loss during network interruption, and high delays due to data flow across the network to the cloud. In addition, most of the implemented parking monitoring systems are locally operated and do not provide real-time information to the driver about where to find a parking space [15]. The utilized communication technologies are based on wired systems or short-range communication methods such as Wi-Fi, Bluetooth, or ZigBee, which are not reliable in most scenarios of indoor and outdoor parking. They also face many issues related to interference, energy consumption, and limited resources.

Moreover, different criteria have been considered for parking-spot selection, such as the preferences of users and operators. The existing systems consider the demand and driver preferences only. The selection decision must be based on a criterion that reduces congestion and maximizes parking utilization and, consequently, revenue. The pricing strategy, on the other hand, in parking lots, is usually based on a fixed rate per hour for short-term parking and a fixed subscription rate for long-term parking, which does not achieve a high revenue. The fixed rate is obtained periodically according to the previous occupancy rate. Such a strategy does not take advantage of the parking availability and demand, and hence it leads to sub-optimal pricing. Therefore, this paper reports on the design and development of a smart parking system based on a multi-layer IoT architecture to resolve the aforementioned issues. The system considers different parking plans, including a short-term parking plan with and without advance reservation and a long-term parking plan. The proposed system is evaluated through implementation and simulation of a selected area representing our case study (i.e., Kudai parking lot). The system has the following features:

- A multi-layer IoT architecture consisting of IoT, fog and cloud layers to overcome the limitations of single gateway-centered solutions. Distributed fog computing nodes are employed to process data locally and forward essential data only to cloud servers. Sensor nodes connect to fog nodes through energy-efficient LoRaWAN communication technology, except for the vision sensors which are connected through Wi-Fi technology. This is because LoRa communication offers a low data rate that is suitable for sensor nodes that do not continuously send large amounts of data.
- A parking-spot selection algorithm that considers arrival time to a parking spot as a parking selection criterion, to reduce waiting time and congestion and, consequently, air pollution. The arrival information for the parking lot, including the rate of entering and departing the parking lot, the parking availability rate, and vehicles' GPS locations are considered. The visual sensor information at junctions is also used to obtain the congestion rate.

- Parking availability estimation that facilitates the parking reservation process with the aim of maximizing parking-lot utilization. Reservation is allowed a minimum of three hours in advance to give walk-in clients the chance of finding a parking spot and, consequently, to reduce traffic congestion at the parking-lot entrance.
- A hybrid pricing algorithm to maximize the parking revenue. The algorithm utilizes an hourly-rate pricing strategy for short-term planning and a dynamic pricing strategy for long-term planning. The dynamic pricing represents a dynamic subscription rate that is set to correspond with demand and revenue.

The rest of the paper is organized as follows. Section 2 discusses related studies on smart parking systems and highlights their limitations and the contributions of our proposed system. In Section 3, the proposed parking system is described, including the proposed multi-layer IoT architecture and the operation workflow, parking-spot selection, parking availability estimation, and hybrid pricing algorithms. Section 4 describes the details of the implementation and simulation evaluation for the case study considered, and the utilized hardware and software tools. The obtained results are also discussed and analyzed. Finally, the conclusions and suggestions for future work are presented in Section 5.

2. Related Work

Several review articles have been presented in the literature that explore, classify, and discuss the technical aspects of smart parking solutions [16–19]. This section discusses related smart parking systems from existing studies, obtained by searching using the keywords ‘smart parking’ and ‘IoT parking’ in several databases such as ISI Web of Science and Scopus. Searching using these keywords resulted in 316 articles, with 127 articles from the ISI Web of Science and 189 articles from Scopus. The duplicated articles from these two databases were removed, resulting in 275 final articles. Figure 1 shows the review methodology used to obtain the related articles considered.

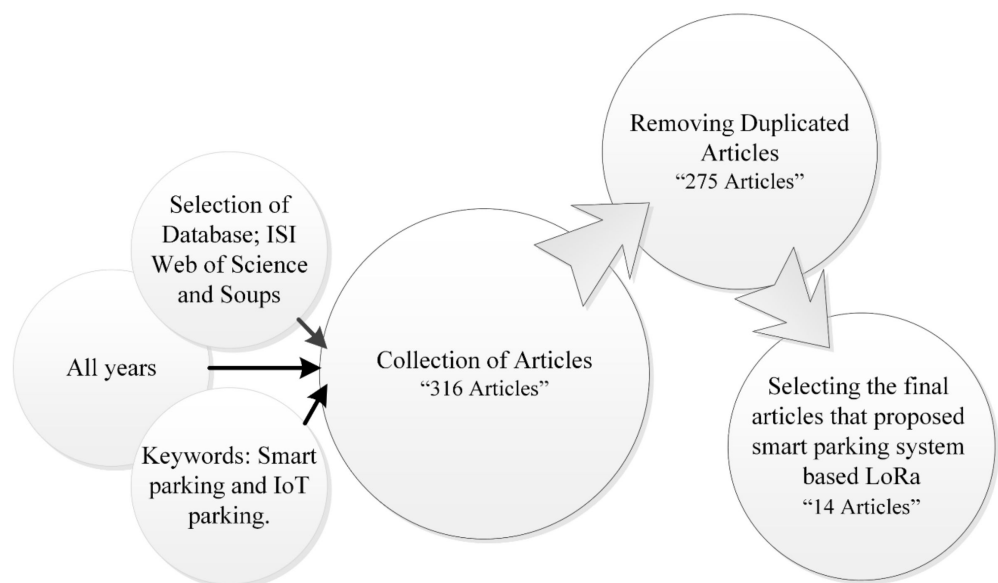


Figure 1. Related articles selection methodology.

The scopes of these articles address different issues, including application and usability [20–22], pricing and contracts [23–25], availability prediction and allocation [26–28], and others. The 275 articles were further scrutinized based on the problem scope, solution approach, and evaluation method. Applying the review methodology helped in extracting the 14 articles most relevant to the scope and objectives of this paper. Different criteria could be considered to classify the systems, including parking environment, provided services, or core sensing technique. Parking systems can be classified based on the core

sensing technique into wireless-sensor-based systems, motion-sensor-based systems, and camera-sensor-based systems. This section will discuss only the smart parking solutions most related to the proposed solution in this article, i.e., wireless-sensor-based systems that utilize LoRa as the network infrastructure of the system.

A proof-of-concept implementation of a smart parking system with LoRa infrastructure has been presented in several papers [29–31]. The developed implementations showed that such a system is efficient in terms of energy, integrity, and scalability for parking monitoring and management in large cities. In [32], the system was implemented with an online reservation system for car parking lots at Universitas Hasanuddin. Similarly, the authors in [33] developed a smart parking system based on LoRa that contains a sensor module, a cloud server module, reservation application, and a third-party payment platform. The system was tested in two cities in Zhejiang province, China: Ningbo and Zhoushan. The experimental results showed an enhancement in monitoring and booking capabilities. However, the quality, security, and reliability of the LoRa communication were not tested in real parking lots in a complex urban environment.

A smart parking solution was proposed in [34] that combines Libelium smart parking sensors, LoRaWAN, and a cluster of Kubernetes with MQTT and MongoDB. The combination enhanced the availability, scalability, and portability of the smart parking system by overcoming the information-sharing bottleneck. The system accumulates all messages in the messaging server for further processing if the main server for information collection fails. Another smart parking solution with a pricing algorithm for revenue maximization was proposed in [35]. The price is dynamically set to balance the available and requested parking spots and ensure the minimum parking fee is charged. The solution comprises multiple sensors that recognize vehicles and obtain contextual and environmental information. Moreover, edge–cloud computing is utilized to minimize the network load and the number of gateways.

Other studies have proposed methods to enhance the interoperability of parking solutions. The system in [36] adopted an IoT gateway-centric architecture [37] that allows real-time interaction between all supported sensor devices and/or actuators through a LoRa gateway using the LoRa network. The gateway, in turn, transforms packets to the format acceptable to the server. Hence, there is no need for a network server to route packets to their destination. The received packets are then parsed by Spring functions and processed with data from the database if required, to finalize the findings and pass them to users. The payment functions for the parking system utilize the API payment highway. In [38], a novel framework-based Wise-IoT system [8] was developed for global interoperability and mobility of IoT applications and devices. The system was developed based on different IoT infrastructures (one was LoRaWAN) and two different interoperability layers. The system was implemented in two cities around the world: Busan and Santander. The generated sensor observations in Santander were sent on different instances of oneM2M and FIWARE IoT platforms, while in Busan, they were distributed using oneM2M as a back end. As a result, the smart parking developed for Santander could be used in a different location in Busan and vice versa.

For privacy preservation in parking systems, the authors in [39] proposed an enhanced LoRaWAN security protocol that provides basic connectivity functions and averts security issues. It prevents a malicious network server from breaking the end-to-end security between a device and its application server. The protocol was examined using different tools and the results showed that the proposed protocol overcomes two handshake options in terms of network latency and signaling overheads. In [40], a new method using a secure element was proposed to safely store keys and process the LoRaWAN protocol. The secure element handles security functions such as data encryption and message signing. In addition, an efficient integrity verification method was proposed for the executable in the main microcontroller unit, based on a secure element. Table 1 summarizes the most relevant related studies.

Table 1. A summary of the most relevant parking systems.

Research Study	Online Platform	Parking Navigation	Reservation Service	Parking Availability Estimation	Payment Platform	Tested Parking Area
Angelov, K.K. 2020.	✓	✓	x	x	x	None
Anshar, M., et al., 2020.	✓	✓	✓	x	x	Campus parking
Shi, J., et al., 2017.	✓	✓	✓	x	✓	City parking
Barriga, J.J., et al., 2020.	✓	✓	✓	x	✓	City parking
Sotres, P., et al., 2018.	✓	✓	x	x	x	City parking
Proposed System	✓	✓	✓	✓	✓	City parking

3. Proposed Smart Parking System

The proposed smart parking system monitors and manages parking lots to make the vehicle parking process more convenient and comfortable by guiding drivers to their parking spot with less congestion and a shorter traveling time. The system consists of cloud computing, fog computing nodes, a group of sensors, and parking management software to acquire and process the data related to parking booking, availability, and pricing. The components have wireless connection capability to send the data to the fog node, which processes the collected data. The system considers three parking classes based on the parking plan: (i) vehicles with a parking subscription, (ii) vehicles with an advance parking reservation, and (iii) vehicles without a parking reservation. The following sub-sections detail the parking system architecture and operation, parking availability estimation, parking-spot selection, and pricing algorithms. Table 2 gives the list of the notations utilized throughout the paper.

Table 2. List of notations.

Symbol	Definition	Symbol	Definition
i	Edge lane	λ_i	Traffic density on that edge
V	Vehicle number in an edge	M_i	Maximum capacity of vehicles on that edge
C	The total congestion rate of all edge lanes on the route	T	The total number of all edge lanes on the route
S_i	A parking spot	S_j	The current state of parking spot S_i
p_i	A parking lot	Cap_i	Capacity of a parking lot
d	Discount percentage	r	Range of hours
P_T	The total parking price	P_F	The first-hour price rate
P_s	The price rate after the first hour	h	Total parking hours
N	The maximum value of ranges	R	The parking revenue
k	The number of available parking spots for subscriptions	P_n	The subscription price rate of n th parking lot.
U_n	The number of clients who afford a specific rate	$Rate_m$	The maximum rate that m th driver affords

3.1. Parking System Architecture and Operation Workflow

A multi-layer IoT architecture for the parking system manages and monitors parking lots efficiently and maintains a low congestion rate. The architecture overcomes the limitations of the traditional sensor–cloud architecture by employing distributed fog computing nodes to process data locally and forward essential data only to cloud servers, instead of forwarding all the data. This maintains fast processing and transmission of the required data only, to achieve a real-time application experience in which the data required for the general view or for long-term services (e.g., parking lots in different locations across a city or historical parking data) are transmitted to the cloud. The architecture consists of three layers: (i) IoT parking layer, (ii) fog layer, and (iii) cloud layer, as shown in Figure 2.

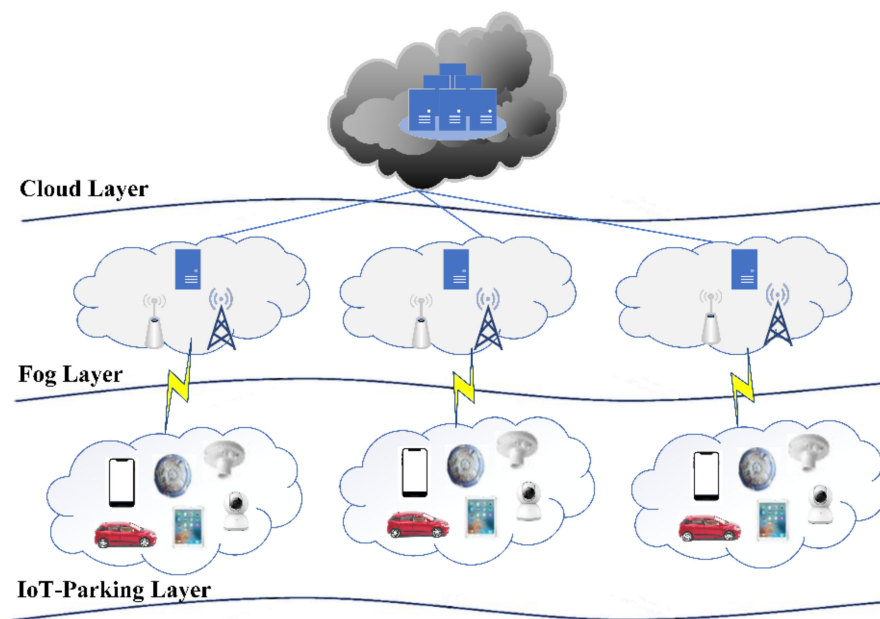


Figure 2. Multi-layer parking system architecture.

3.1.1. IoT Parking Layer

The lowest layer consists of a group of sensor nodes to monitor the parking lots by detecting vehicles and gathering data about the parking area. Different types of sensors could be utilized, placed either in the ground (e.g., electromagnetic sensors), in the ceiling, or in street pillars (e.g., ultrasonic and visual sensors). The most suitable sensor to be utilized is obtained based on the parking lot and its surrounding environment. Magnetometer sensors and thermal cameras are utilized in a parking lot that is exposed to different weather conditions, to detect variations in the magnetic field and differences in temperature. Visual sensors are preferred for more sophisticated parking areas or if they are already installed [41].

In this layer, our system employs several sensors and data acquisition devices including ultrasonic, magnetic, and visual sensors, radio-frequency identification (RFID), Global Positioning System (GPS) equipment, QR code readers and a parking management platform. For instance, visual sensors are placed at the entrance, exits, and junctions to provide useful information for parking-spot selection and to monitor parking rule violations. Solar panels connected to a rechargeable battery are used to supply the power required by the data acquisition devices. Sensor nodes communicate with fog nodes through energy-efficient LoRaWAN communication technology, while vision sensors communicate through Wi-Fi technology. This is because LoRa communication offers a low data rate (i.e., up to 37.5 kbps) which is suitable for sensor nodes that do not continuously send large amounts of data. LoRa technology facilitates long-range communication up to 5 km with 25 mW maximum transmission power [42]. The configuration parameters and frequency carriers of LoRa are set according to the local regulations.

3.1.2. Fog Layer

As mentioned earlier, the purpose of this layer is to provide a robust solution that achieves real-time control and analytics requirements by relocating some of the processing and analysis (e.g., data pre-processing, filtering, and compression) close to the lowest hierarchical layer. This includes the processes of parking-spot selection, parking availability estimation, and pricing in our proposed parking system. The fog layer is usually composed of several fog nodes where the process of a specific service relates to a specific node, according to its requirements. Fog nodes are defined as distributed computing entities that have processing capabilities where each node can be formed by one or more physical

devices (e.g., network device, dedicated server, or computational server). Each node can simultaneously process multiple sensor nodes with the possibility of sensor-node handover if there is a high execution load or a connectivity problem. Moreover, a collaborative task allocation can be performed to share the computational load among fog nodes to reduce the overall operational latency.

3.1.3. Cloud Layer

The highest layer provides virtually unlimited storage and computational resources. The cloud layer facilitates several services for the proposed parking system, such as usage analytics, statistical and historical information, revenue monitoring, and automatic payment and management. The cloud layer also enables secure, real-time online parking reservation information and direction to the reserved parking through a parking platform running on the web or installed in a mobile device. An encryption technique could be employed in the cloud layer to hide the payment plan and driver information, to guarantee security and privacy. Moreover, historical, and long-term information could be used for big data analytics to enhance traffic management and smart city solutions, as well as to enable real-time feedback notification and emergency alerts.

The workflow of the proposed parking system is shown in Figure 3. The process starts by obtaining parking-lot data through sensor nodes and the parking management platform. After data acquisition, the physical variables of the vehicle are sent to the fog nodes, including parking reservation, arrival, and vehicle detection. The fog nodes process the collected data every 10 ms to ensure accurate decisions and minimize the usage of resources and processing capabilities. The acquired data are processed to determine the information on arrival, parking availability, parking matching, and parking reservations. Based on the data, four different algorithms are executed including parking availability estimation, security alerts, dynamic pricing, and parking-spot selection. A detailed description of these algorithms is given in the following sub-sections. The outcome of the algorithms is transmitted to the cloud computing layer through network devices and, consequently, transmits the parking information to the parking management platform.

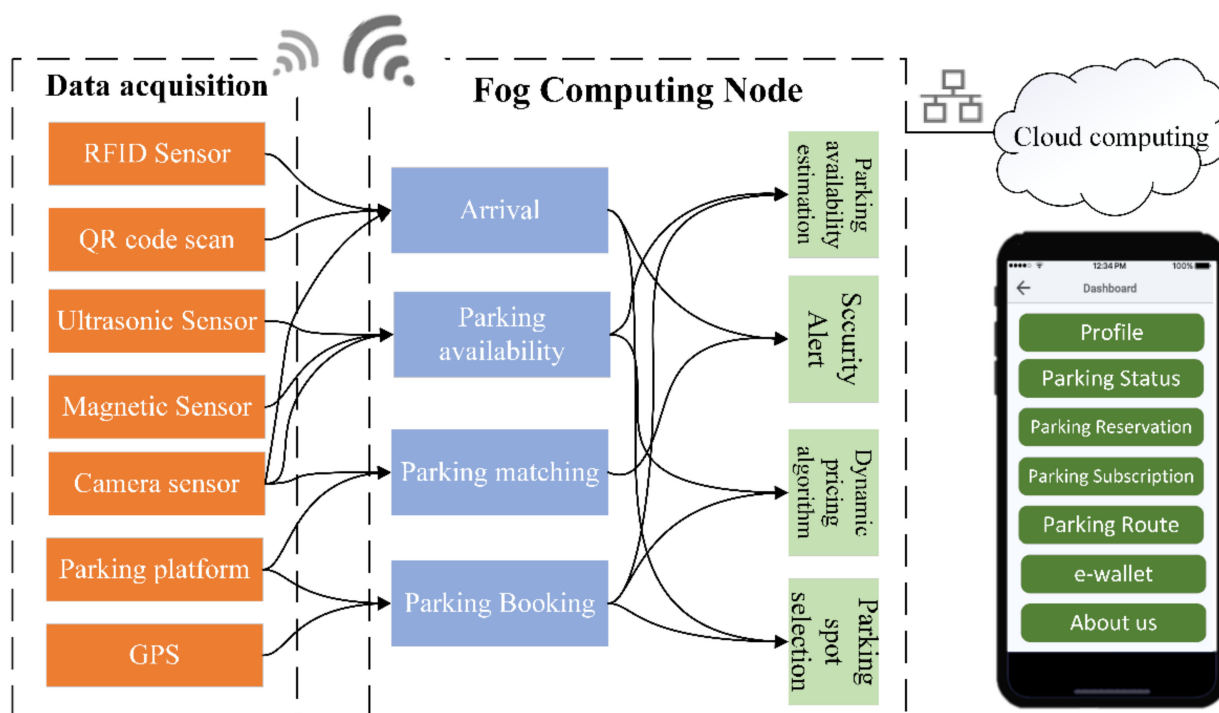


Figure 3. The operation of the smart parking system.

3.2. Parking-Spot Selection Algorithm

The parking-spot selection is initiated upon vehicle entry into the parking lot. The parking-spot selection process uses an optimization algorithm that selects the optimal parking with the aim of minimizing the arrival time to a parking spot. The algorithm is applied to vehicles with and without a parking reservation but not to vehicles with a parking subscription. It utilizes the information on parking arrival and reservation, including the rate of entry and exit, parking availability rate, and the vehicle's GPS location. In addition, the visual sensor information at junctions will be used to obtain the congestion rate using the congestion model presented in [43]. The entrance point is used as a navigation start-up to select the parking lot, which is determined during the vehicle arrival. The algorithm performs a forward search process in which the entrance point is located and then the expected arrival time to an available parking spot is calculated. Moreover, a backtracking process is performed to compute the arrival time from the parking spot back to the preceding entrance point.

An example of the parking-spot selection process is shown in Figure 4. Each road has two edge lanes and a maximum capacity for vehicles. For an edge lane i , the density λ_i of the traffic is calculated as

$$\lambda_i = V / M_i \quad (1)$$

where V represents number of vehicles in the edge and M_i represents the maximum capacity of vehicles in that edge. The density is shown as a percentage in Figure 4. The total congestion rate of all edge lanes on the route is calculated as

$$C = \sum_{i=1}^j \lambda_i / T \quad (2)$$

where T is the total number of all edge lanes on the route. The search for the optimal parking spot starts from a given entrance node to an available parking spot that achieves the minimum travelling time. Accordingly, the search graph consists of a root node which is the entrance point, and the next-level nodes which are all the junctions that can directly be accessed from the previous level (i.e., first root node) and so on, until candidate available parking spots are identified. The graph constitutes a tree-like view in which the maximum depth represents the number of junctions and the candidate parking junction within this depth. The optimal path with minimum travelling time can be found using Dijkstra's algorithm [44].

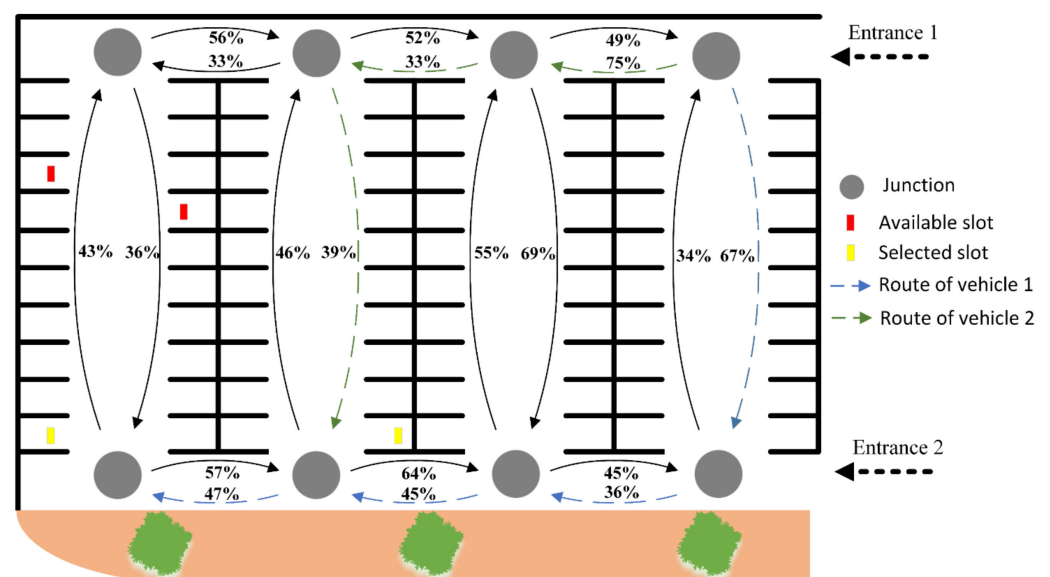


Figure 4. Parking-spot selection and vehicle routing.

3.3. Parking Availability Estimation

Parking availability estimation is proposed to facilitate the parking reservation process. Reservation is allowed a minimum of three hours in advance if it is estimated that parking spots will be available at the requested time. This maximizes parking-lot utilization and gives vehicles without a prior parking reservation the chance to find a parking spot, consequently reducing traffic congestion at the parking-lot entrance. The parking availability at the current instant in time is known by the fog nodes in the system. However, this information is not sufficient for estimating the parking availability in the near future because the vehicle arrival rate fluctuates in different seasonal periods. The traffic fluctuation can be predicted using predictive analytics from the historical seasonal data, which are collected from sensors at parking entrances and junction poles. These historical data are stored in the cloud and categorized according to seasonal periods into several instances. The periods must be small enough (e.g., one minute) to minimize the variations in parking dynamics. These data are used to develop a stochastic model based on a Markov process representing the randomly changing available parking spots.

The Markov model infers the parking availability in a future period from the historical parking availability information in similar periods. The model obtains several parameters for each parking lot including the observation time window, arrival rate, departure rate and state transition matrix. Table 3 shows an example of parking lot data, where multiple historical one-minute-period observations in similar seasons are averaged. The period is set to be short in order to obtain discrete state changes in a parking lot modelled over a time window of five minutes with fixed arrival and departure rates. The reason for choosing a short period is to justify the assumption that arrival and departure rates are fixed within each observation window; however, variations in arrival and departure rates are considered across observation windows. A single state transition matrix that represents the parking dynamics information is obtained for each observation window size.

Table 3. Parking data sample.

Parking Lot p_i				
¹ Season			Period (Time)	Available Spots (Average)
Year	Month	Day		
1439	Safar	5	9 h 56 m	987
1440	Safar	5	9 h 56 m	970
1441	Safar	5	9 h 56 m	950
1442	Safar	5	9 h 56 m	987
1443	Safar	5	9 h 56 m	960

¹ Note: Makkah parking lots form the case study in our work, where the historical parking data fluctuate according to the Hijri calendar; thus, this calendar is used in the system.

For each parking lot, parking states are determined in which each state represents the parking availability range. The future availability state $P(S_i|S_j)$ of a parking spot S_i is obtained using the classic Markov chain model and Kendall's notation, where S_j is the current state [45]. The Poisson distribution is utilized in the simulation to represent the arrival process in periods with a fixed mean value for the number of entering and departing vehicles to/from a parking lot over a single observation period. The proposed parking availability estimation uses five state ranges for a parking lot, from the highly occupied state S_1 to the highly available state S_5 . Figure 5 shows an example of a state transition matrix for a parking lot with capacity $Cap_i = 1000$, which is derived from historical data with a five-minute period of observation. A normalized transition matrix in terms of probabilistic weights is also illustrated in Figure 5.

Algorithm 1: Hybrid pricing algorithm.

BEGIN;

If short-term parking plan is selected **then**

| Process pricing for short-term parking plan;

else if long-term parking plan is selected **then**

| Process pricing for short-term parking plan;

end;

(1) process pricing for short-term parking plan;

Determine the price of the first hour;

Determine the parking hours range;

Calculate the price of the later hours according to the applied discount percentage based on the determined parking hours range;

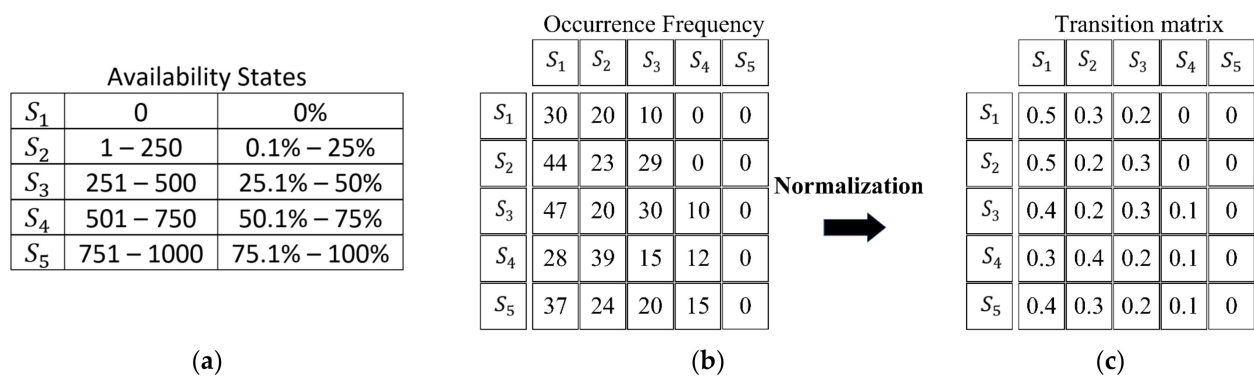
Calculate the total price of the short-term parking which is sum of the price of the first and later hours;

(2) process pricing for long-term parking plan;

Clients bid once or more during selected seasonal periods;

Determine the optimal pricing rate that matches demand and revenue;

Prioritize the clients based on their bidding value and time stamp in a descending order;

**Figure 5.** Markov model: (a) availability states, (b) occurrence frequency, and (c) transition matrix.**3.4. Hybrid Pricing Algorithm**

An algorithm was developed to facilitate the pricing strategy for two types of parking plans: (i) a short-term parking plan (i.e., parking with and without a reservation) and (ii) a long-term parking plan (i.e., monthly and annual parking subscriptions). The hybrid pricing includes an hourly-rate pricing strategy for the short-term plan and a dynamic pricing strategy for the long-term plan. Algorithm 1 shows the pseudocode of the hybrid pricing algorithm.

The hourly pricing rate considers the possibility of specifying certain discount rates. It sets a fixed price rate for the first hour, then the price rate of the later hours is calculated based on ranges in which a specified discount percentage (d) is applied upon exceeding a specified range of hours (r). The total parking price is calculated as follows:

$$P_T = P_F + P_s(h - 1) - XdP_s(h - 1) \quad (3)$$

where P_T is the total parking price, P_F is the first-hour price rate, P_s is the price rate after the first hour, h is total parking hours, and X is calculated as:

$$X = \text{Min}(\lfloor h/r \rfloor, N) \quad (4)$$

where N is the maximum value of ranges. For instance, if parking is required for a total of 20 h, the first-hour parking rate is USD 10, the rate after the first hour is USD 5, and there

is a 10% discount for each of 5 ranges where each range consists of 3 h, the total parking price will be

$$P_T = 10 + (5 * 19) - (5 * 0.1 * 5 * 19) = \$52.5$$

For long-term parking (i.e., subscriptions), a dynamic subscription rate is allocated to correspond with demand and revenue, similarly to [46]. The objective of the strategy is to optimize the parking revenue $R = \sum_{n=1}^k P_n$ where k is the number of parking spots available for subscription and P_n is the subscription price rate of the n th parking lot. This can be achieved by enabling bidding once or more times during selected seasonal periods. The subscription price rate is obtained by firstly allocating the maximum bidding value as the price rate, then reducing it to a rate that matches demand and revenue. For instance, in the scenario where 20 clients bid USD 200 monthly, 3 clients bid USD 150 monthly, and 7 drivers bid USD 90 monthly, the optimal pricing rate will be USD 150 monthly. A maximum bid value will be pre-specified by the parking management to ensure fairness. Upon setting the subscription price rate, the clients who can pay the rate U_n are prioritized based on their bidding value and time stamp in a descending order. The optimization algorithm and maximization objective can be expressed as

$$\begin{aligned} &\text{Maximize } R \\ &\text{Subject to :} \\ &k \leq U_n \\ &\text{Rate}_m \geq P_n ; \quad \forall n \leq k \end{aligned} \tag{5}$$

where Rate_m is the maximum rate that the m th driver will pay. The sorting algorithms Merge Sort or Heapsort [35] can be used to search for the optimal maximum value, due to their low complexity.

4. Implementation and Evaluation

To evaluate the proposed smart parking system, Makkah city parking lots near the Holy Mosque were considered as the case study. Specifically, the parking lot in the Kudai district was selected for simulation and implementation. There are 13 parking lots in Makkah city, with a total area of a million and a half square meters, and they can take up to 45,000 vehicles. Kudai parking lot's area is around 74,000 m² with total of 2500 parking spots. Figure 6 shows parking lots and transportation stations, including Kudai parking lot. The traffic is highly congested during the Hajj and Umrah seasons, but the system is expected to reduce traffic congestion, especially at the parking-lot entrances. According to a general statistical authority, the number of vehicles entering Makkah during the 2019 Hajj season was around 32,978 [47]. The Saudi vision aims to double the total number of pilgrims for the Hajj and Umrah season by 2022, which in turn requires better traffic and parking management.

As a proof of concept, a simplified version of a part of the Kudai parking lot was selected for experimental evaluation, as shown in Figure 7. Note that some of the components are not shown in the figure as they are placed at the back of the parking board. The visual sensor was not included in the proof-of-concept implementation for simplicity purposes. The hardware components of the experiment included HC-SR04 ultrasonic and geomagnetic sensors, an RFID-RC522 reader module, an S50 RFID card, an Adafruit Feather 32u4 with LoRa radio modules, a rechargeable Li-ion battery, a compact photovoltaic panel, LEDs, Raspberry Pi 3, and an HP workstation.

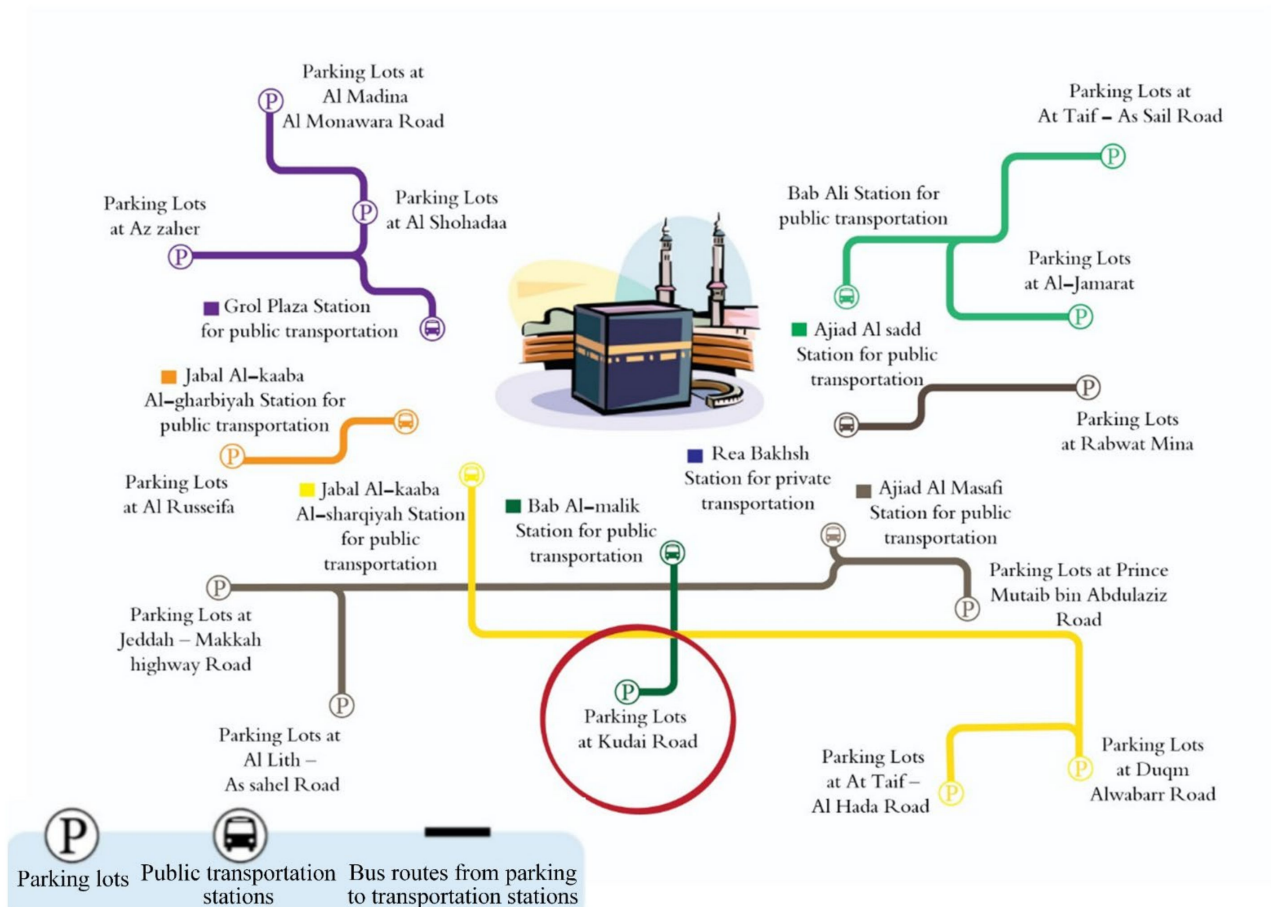


Figure 6. Parking lots and transportation stations near the holy mosque at Makkah city.

A Feather 32u4 LoRa board with an 868/915 MHz radio module was employed for LoRa communication. The serial RX/TX pins of ultrasonic, geomagnetic, and RFID sensors were connected to the Arduino serial RX/TX pins on the Adafruit Feather. Data were read from the Arduino serial RX and transmitted through the radio. The data are received by the onboard LoRa radio and sent through another serial port to the Raspberry Pi, which was used as a fog node in the experiment. The node collects, processes, and sorts the data locally, and forwards the data to the back-end server. It processes parking-spot selection, parking availability estimation, and the hybrid pricing algorithm. A HP workstation running Ubuntu was employed for back-end analysis to monitor and analyze the network's overall performance. It processes the historical data to estimate the parking availability and accordingly allows or does not allow parking reservation. The communication module and the sensors are supplied with power through a rechargeable Li-ion battery that is charged by a compact photovoltaic panel. The Blynk app could be used to display the output of the algorithms executed in the fog node.

A simulation evaluation was conducted to investigate the benefits of the proposed parking system, specifically the slot selection and hybrid pricing algorithms. The free open-source simulator, Simulation of Urban MObility (SUMO) [48] was utilized to define each vehicle explicitly with a specific route in the map. The default behavior of vehicles in SUMO is to drive with highest specified speed with obstacle avoidance. It also contains a parking package to define a parking area and to route vehicles to a specific parking spot [49]. The area of the Kudai parking lot simulated was taken from OpenStreetMap (OSM) and inserted into SUMO to represent the simulation area. Figure 8 shows the Kudai parking lot in the SUMO simulator.



Figure 7. Kudai parking lot and proof-of-concept set-up.

The system was compared with the traditional parking approach in which the driver starts searching for an available parking spot from the current location until one is found. A parking lot with a maximum capacity of 150 parking spots was considered, where 10 and 5 vehicles/minute were the arrival and departure rates, respectively. The initial parking availability was produced randomly in a uniform distribution with a 90% maximum rate. The parking lot was considered full if more than 80% of parking spots were occupied. The measured performance metrics for simulation evaluation were arrival time, searching time, and parking revenue. The arrival time is the time spent by a vehicle in the parking area from the entrance until a selected parking slot is reached, while the searching time is the time spent searching for a parking spot. The parking revenue, on the other hand, is the total revenue from all parking spots with different parking plans in a parking lot.

Figure 9 illustrates the arrival time considering different congestion rates and available parking spots, using the proposed system and the traditional approach. The simulation was executed until an available spot was obtained. Figure 9a shows the arrival time with only one available parking spot, with a congestion rate between 0.1 and 0.9. It shows that the arrival time using the proposed parking-spot selection algorithm was significantly less than for the traditional parking search approach. The traditional search approach consumes around 20% to 60% more time than the proposed algorithm. Although the proposed algorithm always outperformed the traditional approach, the arrival time gap between the two approaches became less with a higher congestion rate because it was difficult for vehicles to move fast at all junctions. In Figure 9b, the arrival time is shown with various available parking spots from 1 to 10 parking spots and a fixed congestion rate of 0.8. The proposed algorithm had an arrival time that was lower than the traditional approach in all cases. When only one parking spot was available, the arrival time of

the proposed algorithm was notably lower compared to the traditional approach which searches blindly for all parking spots in the area until an available spot is found. However, the difference between the arrival times decreased gradually with increasing numbers of parking spots because more available spots could be found.

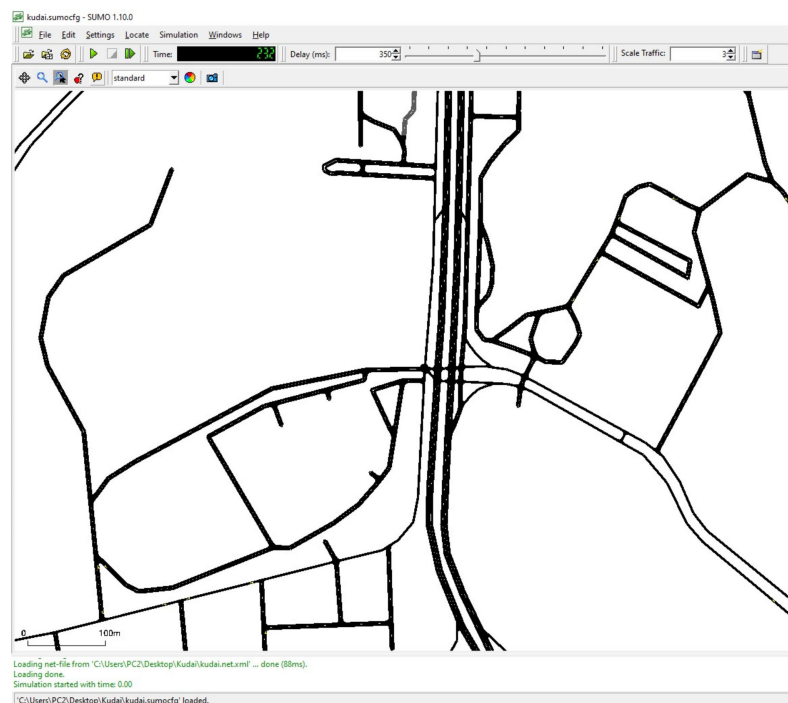


Figure 8. Kudai parking lot in SUMO simulator.

Figure 10 shows the searching time with various available parking spots from 1 to 10 parking spots and a fixed congestion rate of 0.8. The results show that the searching time metric and the arrival time have a similar pattern because the searching time is most of the arrival time period. The proposed algorithm had a lower searching time and arrival time; notably lower when only one parking spot was available. The difference between the two approaches became less when more parking spots were available.

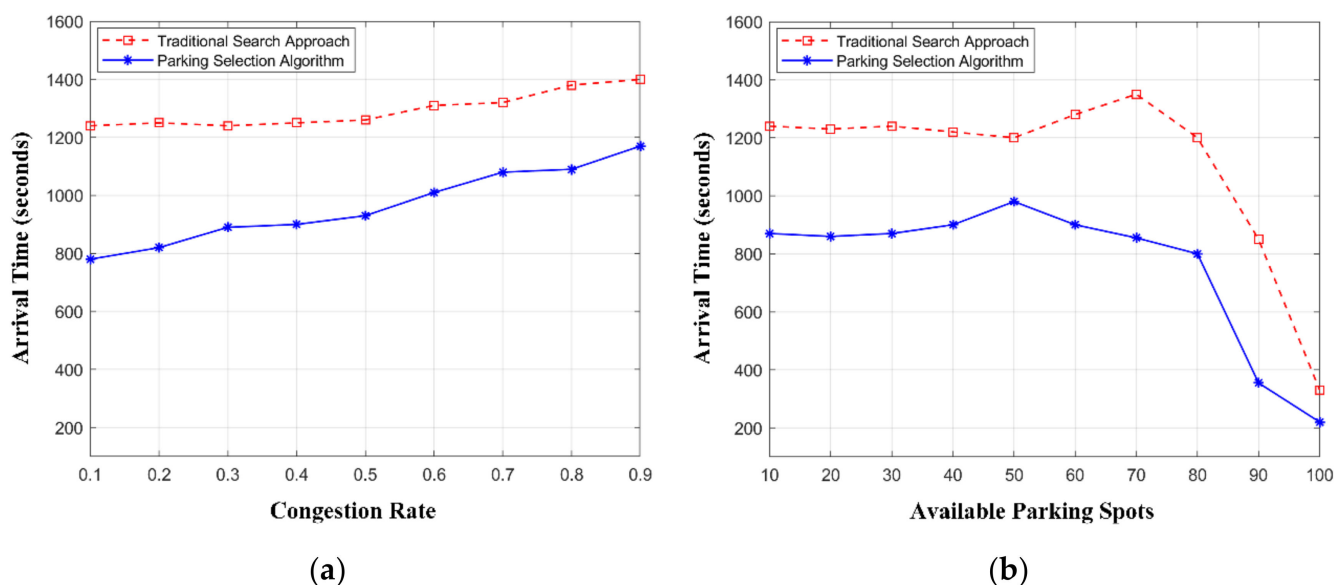


Figure 9. The arrival time using parking selection algorithm and traditional search approach, considering: (a) different available parking spots, and (b) different congestion rates.

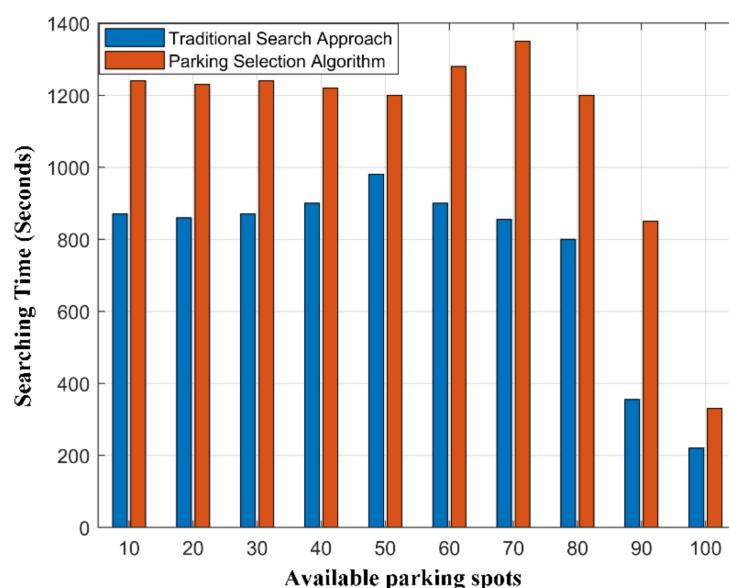


Figure 10. Searching time with different available parking spots using parking selection algorithm and traditional search approach.

The revenue for hybrid and fixed pricing algorithms was evaluated considering different parking reservation requests. Here, a parking lot with a maximum capacity of 150 parking spots was considered, in which 50 spots were allocated for the long-term plan and 100 spots for the short-term plan. Moreover, the number of parking spots available for subscription was considered to be less than the number of the clients who could pay the default subscription rate. The fixed pricing rate was USD 10 per hour in the short-term plan and USD 2000 per month in the long-term plan. For the short-term plan, the dynamic pricing was USD 10 for the first hour with a 10% discount for the remaining hours, while for the long-term plan the bidding rate was randomly higher than the default rate. Figure 11 shows the total revenue from all parking plans, including the short-term and long-term parking plans, compared to fixed pricing. The occupation rate was varied from 10% to 100% for both parking plans. The dynamic pricing scheme increased the revenue notably for the long-term plan, while it was less than the fixed pricing revenue for the short-term plan, due to the discount.

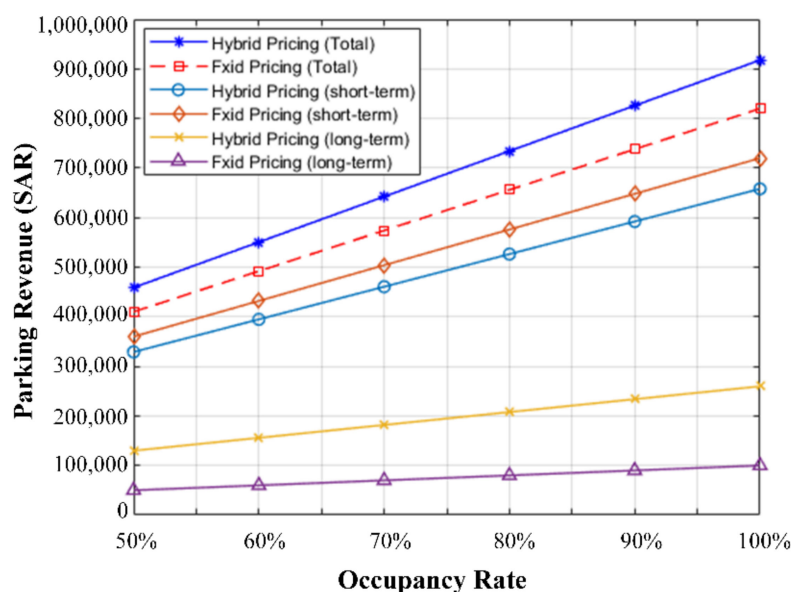


Figure 11. The revenue from the parking lot in one month with different occupation rates.

5. Conclusions

Smart cities aim to improve the quality of life of their residents in different aspects of life including transportation and travel services. In this context, smart parking systems help drivers to obtain and reserve a parking spot in advance. In this paper, a smart parking system was proposed based on a multi-layer IoT architecture with a LoRa communication interface, to overcome the limitations of the current parking systems. The system considered different parking plans including a short-term parking plan with and without advance reservation and a long-term parking plan. The acquired parking data were processed through a fog computing node to facilitate obtaining the required real-time parking data. Moreover, a dynamic pricing algorithm was introduced with aim of maximizing revenue for the parking authority. A novel algorithm to obtain the optimal parking spot with the minimum arrival time was also introduced. The proposed system was evaluated through implementation and simulation of a selected area representing our case study (i.e., Kudai parking lot). The results showed that the parking system enhanced the parking arrival time and revenue. In future work, we plan to develop and install a test field for the proposed parking system in our case study.

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